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Building fault detection data to aid diagnostic algorithm creation and performance testing

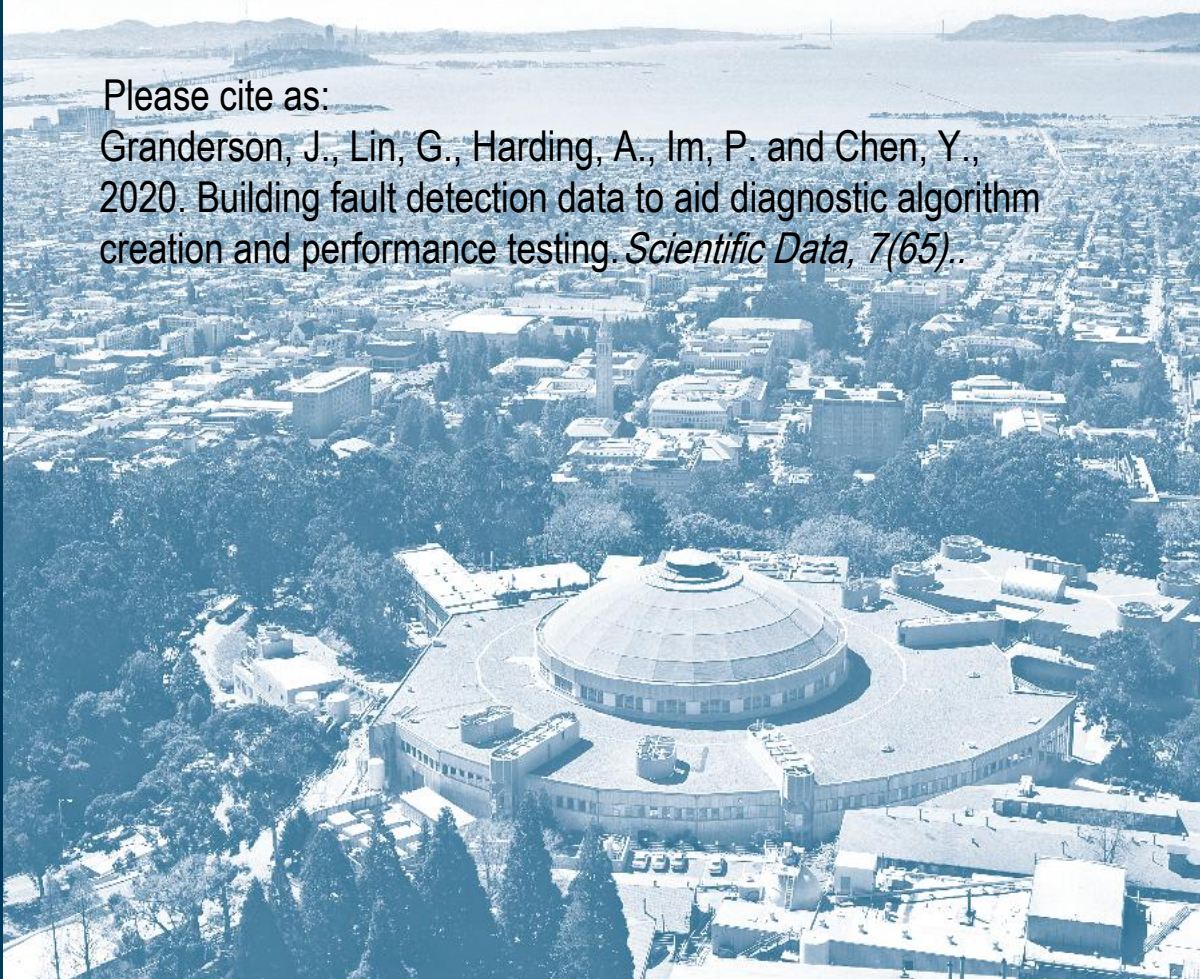
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Building fault detection data to aid diagnostic algorithm creation and performance testing

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Abstract

It is estimated that approximately 4-5% of national energy consumption can be saved through corrections to existing commercial building controls infrastructure and resulting improvements to efficiency. Correspondingly, automated fault detection and diagnostics (FDD) algorithms are designed to identify the presence of operational faults and their root causes. A diversity of techniques is used for FDD spanning physical models, black box, and rule-based approaches. A persistent challenge has been the lack of common datasets and test methods to benchmark their performance accuracy.

This article presents a first of its kind public dataset with ground-truth data on the presence and absence of building faults. This dataset spans a range of seasons and operational conditions and encompasses multiple building system types. It contains information on fault severity, as well as data points reflective of the measurements in building control systems that FDD algorithms typically have access to. The data were created using simulation models as well as experimental test facilities, and will be expanded over time.

Background & Summary

Buildings use 40% of primary energy globally, and account for 33% of direct and indirect carbon emissions from fuel combustion¹. In US commercial buildings, an average 29% of energy use can be reduced through more efficiency operations and improved controls². Algorithms developed to perform automated fault detection and diagnostics (FDD) use building operational data to identify the presence of faults and (in some cases) isolate their root causes.

As buildings become more data rich, and as data science comes to buildings, FDD is of increasing relevance to the building community. Outside of the research community, building owners and operators at the leading edge of technology adoption are using AFDD to enable median whole-building portfolio savings of 7%³. Modern commercial and research-grade FDD technologies often integrate with building automation systems (BAS) to obtain operational controls data for their algorithms, or are implemented as retrofit add-ons to existing equipment. Extensive libraries of detection logic are continuously run against the data, and results are surfaced through a graphical user interface for resolution by operations and maintenance staff⁴.

A diversity of techniques is used for FDD in buildings, spanning physical models, black box, and rule-based approaches and researchers continuously strive to develop new and better algorithms, with hundreds of methods published in the literature⁵. A persistent challenge has

been the lack of common datasets and test methods to support the development of, and to benchmark the performance accuracy of FDD methods against one another. Prior work has made progress toward common test methods^{6,7}, however test datasets remain a gap.

Overall, there are limited examples of publicly available *operational* datasets for building energy efficiency applications. For example, utility smart meter data, HVAC control system data, lighting system data, and submetered electricity and gas data are often obtained on a research-project specific data, and restricted by NDAs or other data sharing restrictions. There is a nascent body of shared operational datasets for buildings, including for example^{8,9,10}.

Specific to FDD applications, it is extremely rare to find datasets that have verified ground truth information on the presence and absence of faults. The majority of buildings have not yet implemented FDD, and often faults go undetected. Where FDD tools *have* been implemented, the historians do not indicate whether detected faults were verified, and false positives and negatives may confound interpretation of the historic records. While BAS are common in larger buildings, and may contain time series trend logs of operational data in their historians, these data are not labeled to indicate whether it represents faulted, un-faulted, or simply atypical/anomalous system operational states.

The dataset described in this article contains operational building heating ventilation and air-conditioning (HVAC) data, paired with validated ground-truth information as to the presence and absence of faults. This dataset spans a range of seasons and operational conditions and encompasses multiple building system types, fault types, and fault severity, or intensity levels. The systems of focus include air-handling units (AHUs) and rooftop units (RTUs). The included data points reflect measurements that are typically logged in building control systems. The dataset comprises both simulated (i.e., modeled) data, and experimental (i.e., physical) data from test facilities. The data were provided by multiple contributors, and synthesized into a single repository with a common format and documentation.

The test dataset can be used by FDD developers, FDD users, and research funders to:

- Compare and contrast performance accuracy across FDD algorithms
- Identify performance gaps to focus future development efforts and resource investment
- Develop an understanding of how FDD technology overall is improving over time

A preliminary illustration of use of the dataset to compare and contrast FDD algorithm performance accuracy and identify performance gaps is documented in¹¹. This initial dataset will be expanded over time to cover a larger range of operational conditions, fault types, and seasons. It will also be evolved to include a larger set of HVAC systems, specifically, chiller and boiler plants, dual-duct AHUs, terminal variable air volume (VAV) boxes, and terminal fan coil units.

Methods

The dataset¹² comprises 5 AHU and RTU HVAC system types, created either through simulation, or in physical experimental facilities by multiple contributors. In the following sections we describe: the tools and facilities used to create the data, system configurations and control sequences, fault profiles (type, intensities and durations), and methods of fault imposition.

Facilities and simulation tools

The simulated datasets were created using HVACSIM+ and an EnergyPlus-Modelica co-simulation. HVACSIM+ was developed by the US National Institute for Standards and Technology¹³, the Modelica Buildings Library¹⁴ is developed by the Lawrence Berkeley National Laboratory, and EnergyPlus¹⁵ is developed by several contributors through funding from the US Department of Energy. Described with respect to other modeling tools in¹⁶, HVACSIM+, Modelica, and EnergyPlus are non-proprietary tools to model the behavior of building HVAC systems using physics-based approaches.

The experimental datasets were created using three experimental research facilities. Located at the Lawrence Berkeley National Laboratory in Berkeley, California, FLEXLAB[®] ¹⁷ is designed to evaluate the efficacy of major building systems, individually or as an integrated whole, under real-world conditions. FLEXLAB testbeds can monitor and assess heating, ventilation, air conditioning, lighting, windows, building envelope, control systems and plug loads in any combination. Each building in the facility features two identical paired test cells. The facility is operated through a National Instruments control and data acquisition platform. The test cell used to create the dataset described in this paper comprised a 20- by 25-foot (6.1- by 7.6-meter) zone served by a 10-ton (35.2-ton) direct expansion chiller shared with the adjacent cell. The test cell contains a dedicated air-handling unit with water-sourced heating and cooling coils and a direct-drive variable frequency drive (VFD) controlled fan. Heating water is provided by a natural gas boiler. The test cell features a south-facing windowed wall, insulated concrete slab, and otherwise near-adiabatic walls and roof.

Two experimental data sets were taken in FLEXLAB, one in single-zone constant air volume (CAV) mode and one based on ASHRAE Guideline 36¹⁸ single-zone variable air volume (VAV) mode. Both were implemented in National Instruments' Test Stand programming environment. The CAV mode included a modulating, staged economizer for cooling mode.

Located at the Oak Ridge National Laboratory in Oak Ridge Tennessee, the Flexible Research Platform (FRP)¹⁹ is a two-story building with 10 conditioned zones and 2 unconditioned zones (i.e., stair case). The building is 13.4 x 13.4m with a 40.6cm thick exterior wall. The 10 conditioned zones comprise eight perimeters and two cores areas. The multi-zone HVAC system used for the data described in this paper incorporates a 44kW RTU and a natural gas furnace.. Each room in the FRP has a variable-air-volume (VAV) box with electric resistance reheat. The central fan in the air-handling unit draws return air from each room. To create the data, the original intake for the fresh air in the RTU was blocked (i.e., no ventilation air) and an exhaust fan (with a known air flow rate) was left un-operated. The facility uses a dedicated Johnson Controls Metasys building automation system, through which the room set point temperatures, schedules, and other controls were implemented.

Located at the Iowa Energy Center in Ames City, Iowa, the Energy Resource Station facility was built to compare different energy efficiency measures and monitor their energy consumption and performance. The test system was controlled by a commercially available building automation system. The fault tests were conducted on an AHU serving three perimeters, and one interior zone. The system was configured to provide variable air volume space conditioning. The chilled water is provided from an air-cooled chiller and the heating water is provided by a natural gas-fired boiler. More detailed information about this facility is provided in²⁰.

System configurations and control sequences

Single-zone constant air volume (CAV) and variable air volume (VAV) AHU

Figure 1 contains the schematic representation of the single-zone AHU.

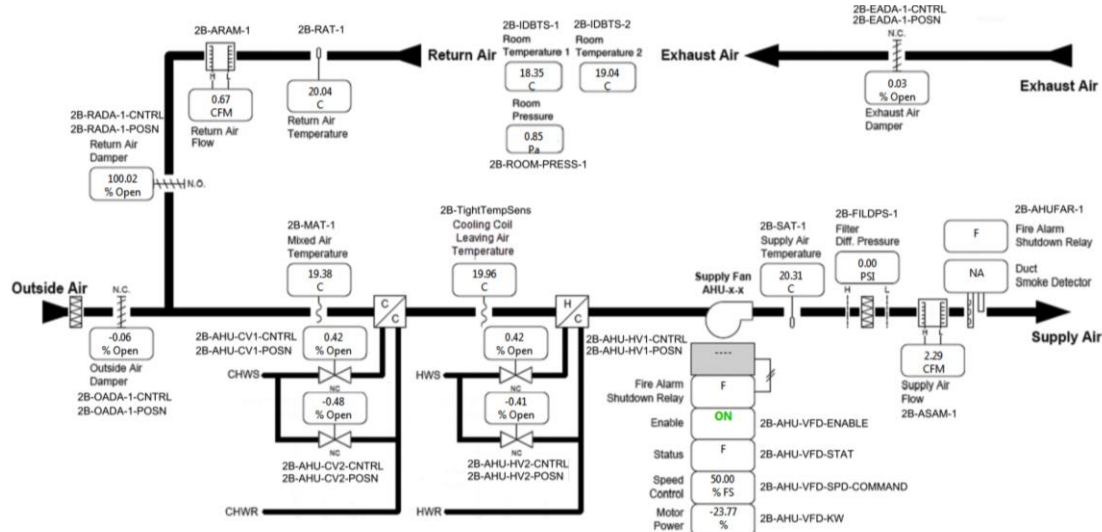


Figure 1. Schematic diagram of the single-zone AHU.

Control sequence for the CAV configuration: The AHU is scheduled for automatic operation on a time of day basis for occupied and unoccupied mode. There is no dehumidification control.

Occupied mode (Monday-Sunday 6am-6pm)

- Fan status: The supply fan continues to run.
- Supply air temperature control: the cooling coil valve and heating coil valve shall modulate to maintain a SAT setpoint. The SAT setpoint is reset within T_{min} (10°C) and T_{max} (30°C) based on zone demand.
- Supply air fan speed control: The supply fan speed is fixed at 50%.
- Minimum outdoor air control: When the unit is not in economizer mode, the OA damper shall be fixed at a minimum OA damper position (15% opening).
- Economizer mode: The AHU shall enter economizer mode when outdoor air temperature is 2°C lower than the return air temperature. The OA damper will gradually open to 100%, then RA damper will gradually close to 0% and EA damper will gradually open to 100%.
- Space temperature control: The zone heating and cooling setpoint are 21.7°C and 23.3°C during the occupied time period.

Unoccupied mode

- The supply fan is on. The OA and EA damper close and the RA damper fully open.
- Unoccupied heating: zone air temperature heating setpoint is 18.3°C.
- Unoccupied cooling: zone air temperature cooling setpoint is 26.7°C.

Control sequence for the VAV configuration: This was an advanced sequence modified from, and based upon the ASHRAE Guideline 36. The AHU was scheduled for automatic operation on a time of day basis for occupied and unoccupied mode. There was no dehumidification control.

181 Occupied mode (Monday-Sunday 6am-6pm)

- 182 • Fan status: The supply fan continues to run.
- 183 • Supply air temperature control: In cooling mode, the heating coil valve is closed
- 184 and the cooling coil valve shall modulate to maintain a SAT setpoint. The SAT
- 185 cooling setpoint is reset within T_min (12.8°C) and T_max (22.5°C) based on zone
- 186 demand; In heating mode, the cooling coil valve is closed and the heating coil valve
- 187 shall modulate to maintain a supply air temperature (SAT) setpoint. The SAT
- 188 cooling setpoint is reset within T_min (22.5°C) and T_max (30°C) based on zone
- 189 demand. In economizer mode, the OA damper shall modulate to maintain the SAT
- 190 heating setpoint.
- 191 • Supply air fan speed control: The supply fan speed is reset between minimum
- 192 (10%) and maximum speed (50% in cooling mode, 30% in heating mode) based on
- 193 zone demand. The minimum speed is determined to meet the ventilation with the
- 194 OA damper completely open. The maximum speed is determined to provide
- 195 design heating/cooling airflow for heating/cooling mode.
- 196 • Minimum outdoor air control: When the unit is not in economizer mode, the OA
- 197 damper shall be fixed at a minimum OA damper position which is reset based on
- 198 supply fan speed between minimum (10%) and maximum (15%). Return air
- 199 damper is fully open and exhaust air damper is fully closed.
- 200 • Economizer mode: The AHU shall enter economizer mode when outdoor air
- 201 temperature is 2°C lower than the return air temperature. The OA damper will
- 202 open to 100%, while RA damper will gradually close to 0% and EA damper will
- 203 gradually open to 100%.
- 204 • Space temperature control: The zone heating and cooling setpoint are 21.7°C and
- 205 23.3°C during the occupied time period.

206 Unoccupied mode

- 207 • The supply fan run at minimum speed (10%). The system operates in the same
- 208 way as in occupied mode to when the space temperature beyond the unoccupied
- 209 heating/cooling setpoint, and disabled when the setpoint +/- 2°C is achieved.
- 210 • Unoccupied heating: zone air temperature heating setpoint is 18.3°C.
- 211 • Unoccupied cooling: zone air temperature cooling setpoint is 26.7°C.

212 *Multi-zone VAV AHU #1*

213 Figure 2 contains the schematic representation of the multi-zone VAV AHU #1.

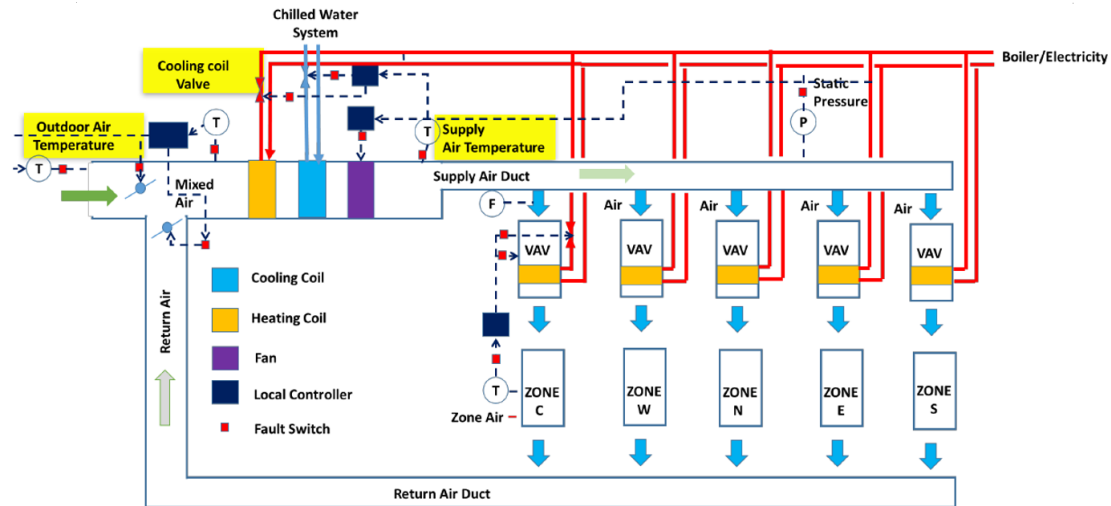


Figure 2: Schematic diagram of multi-zone VAV AHU #1.

Control sequence: The control sequence was modified from, and based on ASHRAE 90.1-1989²¹ and 1999²². The AHU was scheduled for automatic operation on a time of day basis for occupied and unoccupied mode. There was no dehumidification control.

Occupied mode (Monday – Saturday 6am – 8pm)

- Fan status: The supply fan and return fan starts or continue to run.
- Supply air temperature control: The cooling coil valve shall modulate to maintain a fixed 12.7°C supply air temperature setpoint.
- Static pressure control: The supply fan VFD shall modulate to maintain a fixed 250pa static pressure set point. Return fan VFD is controlled as the same as supply fan.
- Minimum outdoor air control: When the unit is not in economizer mode, the OA damper shall be fixed at a minimum OA damper position (14% opening)
- Economizer mode: The AHU shall enter economizer mode when outdoor air temperature is below 15.5°C. The OA damper will modulate in sequence with return air damper to maintain the supply air temperature setpoint. The cooling coil valve will be closed. Once the OA damper is greater than 100% open. The cooling coil valve shall be enable to maintain supply air temperature setpoint
- VAV box reheating coil valve and airflow control: In cooling mode, when the zone cooling setpoint is met, VAV airflow is 30% of max flow rate; when the zone temperature is -1.7°C higher than the setpoint, the damper is 100% open or max airflow rate, when zone temperature is between the setpoint and setpoint +1.7°C, the damper modulate so that the airflow rate is 30% and 100% of max flow rate. In heating mode, VAV airflow is 30% of max flow rate, when the zone temperature is -1.7°C lower than the setpoint, the heating coil valve is fully open; when the zone temperature is at zone temperature heating setpoint, the heating coil valve is 0% open. When zone temperature is between the heating setpoint and setpoint -1.7°C, the heating coil modulate between 0% to 100%. When the zone temperature is between zone cooling setpoint and heating setpoint, VAV airflow is 30% of max flow rate.

- Space temperature control: The zone heating and cooling setpoint are 21.1°C and 23.8°C during the occupied time period.

Unoccupied mode

- Fan status: The supply fan is off. The cooling coil valve closes and the OA damper close. The return fan is controlled as the same as supply fan. System cycling ON and OFF to maintain the unoccupied heating and cooling setpoint.
- Unoccupied heating: zone air temperature heating setpoint is 15.5°C.
- Unoccupied cooling: zone air temperature cooling setpoint is 26.6°C

Multi-zone VAV AHU #2

Figure 3 contains the schematic representation of the Multi-zone AHU #2. An experimental dataset and a simulated data set were created based on Multi-zone AHU #2.

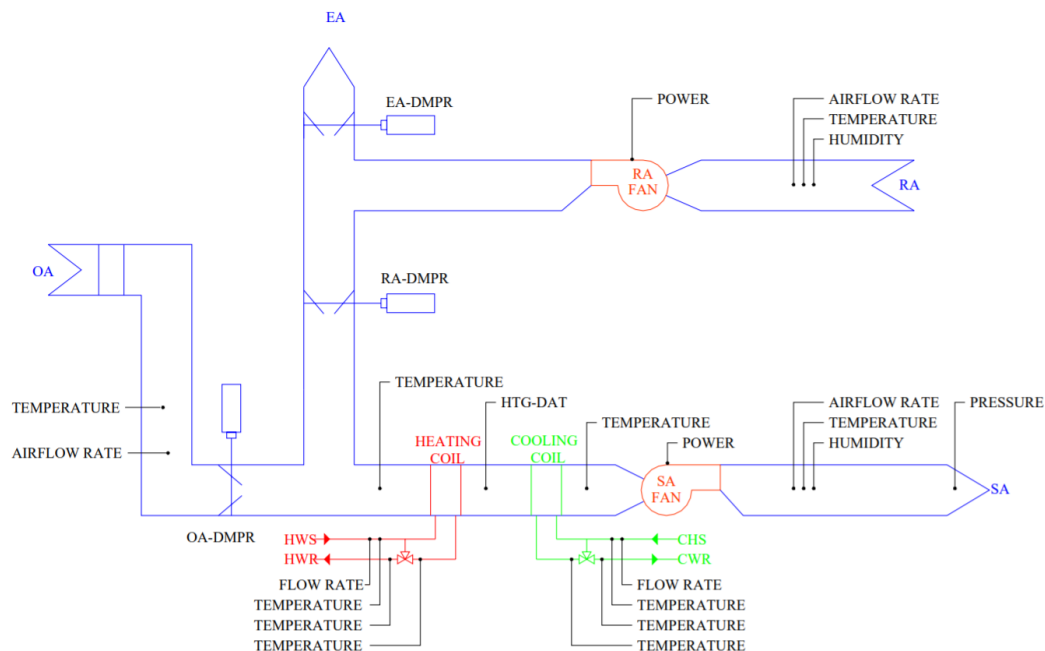


Figure 3: Schematic diagram of multi-zone VAV AHU #2.

Control sequence: The AHU is scheduled for automatic operation on a time of day basis for occupied and unoccupied mode. There is no dehumidification control.

Occupied mode (Monday - Sunday 6am -6pm)

- Fan status: The supply fan and return fans start or continue to run.
- Supply air temperature control: The cooling coil valve shall modulate to maintain a fixed 12.7°C supply air temperature setpoint. When the outdoor air damper is in the minimum position and mechanical heating is required (output from the PI control algorithm drops below 0) causing the control sequence to switch to the mechanical heating mode. During the mechanical heating mode, the valve for the AHU heating coil is modulated to maintain a fixed 18.3°C supply air temperature setpoint.

- Static pressure control: The supply fan VFD shall modulate to maintain a fixed 9653Pa static pressure set point. The return fan is operated with a speed tracking control sequence (80% of supply fan speed).
- Minimum outdoor air control: When the unit is not in economizer mode, the OA damper shall be fixed at a minimum OA damper position (40% opening)
- Economizer mode: The AHU shall enter economizer mode when outdoor air temperature is below 18.3°C. The OA damper will modulate in sequence with return air damper to maintain the supply air temperature setpoint. The cooling coil valve will be closed. Once the OA damper is greater than 100% open. The cooling coil valve shall be enable to maintain supply air temperature setpoint
- VAV box reheating coil valve and airflow control: If zone temperature is less than zone heating setpoint, then a heating case exists. The VAV damper is regulated to maintain a minimum air flow rate, determined either for indoor air quality or equipment limitations. The reheating valve is regulated by a dual PI (DPI) algorithm to supply enough heated water flowing through the reheating coil to increase the entering air temperature to bring zone air temperature above heating setpoint. When zone air temperature is higher than the zone cooling setpoint, then a cooling case exists. The reheating valve position is at 0 %. The VAV damper is opened to bring in more supply air to cool the zone. The air flow rate entering the zone may be varied between the minimum value and the maximum value, which is the rated maximum flow rate for the VAV unit. An air flow rate setpoint is determined by scaling the DPI output between minimum and maximum values. Another PI then regulates the damper position to maintain air flow rate setpoints. Maximum air flow rate is 1699 cubic m/hr for exterior zones and 680 cubic m/hr for interior zones. Minimum air flow rate is 340 cubic m/hr for all zones
- Space temperature control: The zone heating and cooling setpoint are 21.1°C and 22.2°C during the occupied time period.

Unoccupied mode

- The fans are turned off, and the dampers and valves are indexed to a fully closed position. Fully closed dampers and valves refers to 100 % return air with both the heated and chilled water valves closed.

Rooftop unit

Figure 4 contains the schematic representation of the RTU.

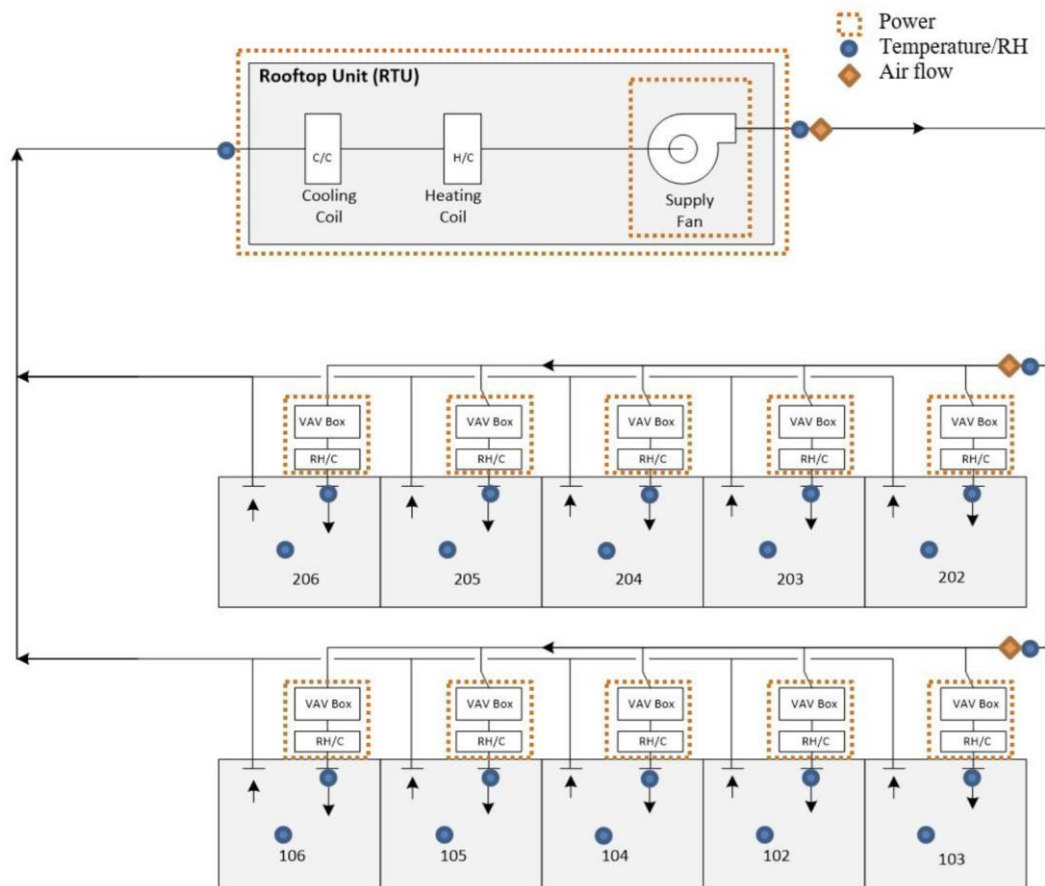


Figure 4. Schematic diagram of the RTU and connected 10 VAV boxes serving 10 zones.

Control sequence for the RTU: The AHU is scheduled for automatic operation on a time of day basis for occupied and unoccupied mode.

Occupied mode (Monday-Sunday 7am-10pm)

- Supply air temperature control: the cooling coil valve and heating coil valve shall modulate to maintain a SAT setpoint. The SAT setpoint is 12.7°C year-round
- Space temperature control: The zone heating and cooling setpoint are 21°C and 24°C during the occupied time period
- The exception was for the condenser fouling test, for which occupied mode started at 6:38am and ended at 9:38pm

Unoccupied mode

- Unoccupied heating: zone air temperature heating setpoint is 15.5°C
- Unoccupied cooling: zone air temperature cooling setpoint is 26.6°C

Fault profiles

Table 1 summarizes the faulted and unfaulted scenarios for the AHU data, including the fault type, intensity, and season in which data were acquired. Each cell contains the number of days for which data were acquired for each scenario. Table 2 contains the same information for the RTU data.

Input Scenarios			MZVAV AHU-1 (Sim)				MZVAV AHU-2 (Exp)		MZVAV AHU-2 (Sim)			SZCAV AHU (Exp)	SZVAV AHU (Exp)
Fault Type		Fault Intensity	Sp	Sum	F	W	Sp	Sum	Sp	Sum	W	W	Sum
OA Damper	Stuck	Min position							1		1		1
		Fully open										1	1
		40% open							1				
		45% open								1			
		50% open								1		1	
Valve of Heating Coil	Stuck	Fully Closed										1	
		50% open										1	1
		Fully Open										1	1
	Leaking	Low						1		1		1	
		Medium						1		1			
		High						1		1		1	1
Valve of Cooling Coil	Stuck	Fully Closed							1			1	
		Fully Open							1	1		1	1
		15% Open								1			
		50% Open										1	
		65% Open								1			
	Leaking	Low										1	
		High										1	1
Outdoor Air Temperature	Bias	+1°C	7	7	7	7							
		+2°C	7	7	7	7							
		+4°C	7	7	7	7							
		-1°C	7	7	7	7							
		-2°C	7	7	7	7							
		-4°C	7	7	7	7							
Unfaulted			7	7	7	7	4	3	3	9	1	1	4

Table 1. Fault profiles for the AHU data; Sim = Simulated, Exp = Experimental, Sp = Spring, Sum = Summer, F = Fall, W = Winter.

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Input scenarios		Season		
Fault type	Fault intensity	Sum	W	F
Condenser fouling	25% reduction in condenser coil air flow full load	1		
	50% reduction in condenser coil air flow full load	1		
HVAC setback error: delayed onset	3-hour onset delay		1	
HAV setback error: early termination	3-hour early termination		1	
Excessive infiltration	+20% infiltration		1	
	+40% infiltration		1	
Lighting Setback Error: Delayed Onset	3-hour onset delay		1	
Lighting Setback Error: Early Termination	3-hour early termination		1	
No Overnight HVAC Setback	No setback		1	
No Overnight Lighting Setback	No setback		1	
Thermostat measurement bias	Bias of +2.2°C (Core zone 103)		1	
	Bias of -2.2C (Core zone 103)		1	
	Bias of +2.2(Perimeter zone 205)		1	
	Bias of -2.2(Perimeter zone 205)		1	
Unfaulted			6	1

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Table 2. Fault profiles for the RTU data; Sum = Summer, F = Fall, W = Winter.

Methods of fault imposition

Tables 3 through 8 summarize how each fault was imposed for each of the represented systems and fault scenarios.

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Input Scenarios			Method of fault imposition
Fault type		Fault intensity	
OA damper	Stuck	Fully open (100%)	Automated override of control signal values to indicate that OA damper is stuck.
		Partially open (50%)	
	Leaking	20% of max damper flow	If control signal drops below X%, fix control output at X%. Otherwise damper controls normally. X = 2 at 20% intensity, and = 10 at 50% intensity
		50% of max damper flow	
Valve of Heating Coil	Stuck	Fully closed (0%)	Automated override of control signal values to indicate that heating coil valve is stuck.
		Fully open (100%)	
		Partially open (50%)	
	Leaking	5% of max coil valve flow	Open heating coil bypass valve to 5%/40% of the maximum heating coil valve flow.
		40% of max coil valve flow	
Valve of Cooling Coil	Stuck	Fully closed (0%)	Automated override of control signal values to indicate that cooling coil valve is stuck
		Fully open (100%)	
		Partially open (50%)	
	Leaking	5% of max coil valve flow	Open heating coil bypass valve to 5%/50% of the maximum heating coil valve flow.
		50% of max coil valve flow	
Unfaulted			--

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Table 3. Methods of fault imposition for each of the single-zone CAV AHU faults.

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Input Scenarios			Method of fault imposition
Fault type		Fault intensity	
OA Damper	Stuck	Minimum position	Automated override of control signal values to indicate that OA damper is stuck.
		Fully open (100%)	
Valve of Heating Coil	Stuck	Fully open (100%)	Automated override of control signal values to indicate that heating coil valve is stuck.
		Partially open (50%)	
	Leaking	40% of max coil valve flow	Open heating coil bypass valve to 40% of the maximum heating coil valve flow.
Valve of Cooling Coil	Stuck	Fully open (100%)	Automated override of control signal values to indicate that cooling coil valve is stuck
	Leaking	50% of max coil valve flow	Open heating coil bypass valve to 50% of the maximum heating coil valve flow.
Unfaulted			--

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352 Table 4. Methods of fault imposition for each of the single-zone VAV AHU faults.

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Input Scenarios	
Fault type	Fault intensity
Outdoor air temperature sensor bias (x is the true value, x' is the faulted value)	$x' = x + 1(^{\circ}\text{C})$
	$x' = x + 2(^{\circ}\text{C})$
	$x' = x + 4(^{\circ}\text{C})$
	$x' = x - 1(^{\circ}\text{C})$
	$x' = x - 2(^{\circ}\text{C})$
	$x' = x - 4(^{\circ}\text{C})$
Unfaulted	-

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Table 5. Methods of fault imposition for each of the multi-zone VAV AHU #1 faults.

Input Scenarios			Method of fault imposition
Fault type		Fault intensity	
Valve of Heating Coil	Leaking	Stage 1: 1.5 SLM (Standard liter per minute)	Manually open heating coil bypass valve
		Stage 2: 3.7 SLM	
		Stage 3: 7.5 SLM	
Unfaulted			-

356 Table 6. Methods of fault imposition for each of the multi-zone VAV AHU #2.1 faults.

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Input Scenarios			Method of fault imposition
Fault type		Fault intensity	
OA Damper	Stuck	Fully closed	Automated override of control signal values to indicate that OA damper is stuck.
		40% open	
		45% open	
		55% open	
Valve of Heating Coil	Leaking	Stage 1: 1.5 SLM	Manually open heating coil bypass valve
		Stage 2: 3.7 SLM	
		Stage 3: 7.5 SLM	
Valve of Cooling Coil	Stuck	Fully closed	Automated override of control signal values to indicate that cooing coil valve is stuck.
		Fully open	
		Partially open 15%	
		Partially open 65%	
Unfaulted			-

358 Table 7. Methods of fault imposition for each of the multi-zone VAV AHU #2.2 faults.

Input Scenarios		Method of fault imposition
Fault type	Fault intensity	
Condenser Fouling	25% reduction in condenser coil air flow full load	Cover the condenser face using screen, mesh, or cloth
	50% reduction in condenser coil air flow full load	
HVAC Setback Error: Delayed Onset	3-hour onset delay	Modify the control programming
HVAC Setback Error: Early Termination	3-hour early termination	Modify the control programming
Excessive infiltration	+20% infiltration	Open windows to achieve target infiltration area
	+40% infiltration	
Lighting Setback Error: Delayed Onset	3-hour onset delay	Modify the control programming
Lighting Setback Error: Early Termination	3-hour early termination	Modify the control programming
No Overnight HVAC Setback	No setback	Modify the control programming
No Overnight Lighting Setback	No setback	Modify the control programming
Thermostat measurement bias	Bias of +2.2°C (Core zone 103)	Adjust the temperature set point
	Bias of -2.2°C (Core zone 103)	
	Bias of +2.2°C (Perimeter zone 205)	
	Bias of -2.2°C (Perimeter zone 205)	
Unfaulted		-

360 Table 8. Methods of fault imposition for each of the rooftop unit faults.

361 **Code availability**

The Modelica Buildings Library and EnergyPlus are freely available for download^{23,24}. EnergyPlus runs on Windows, Mac OSX, and Linux operating systems. A Windows or Linux-based computer and Dymola solver are required to run Modelica, and Dymola can be licensed from Modelica Buildings Library. HVACSim+ is also freely available, upon request from NIST, and has no operating system requirements. The HVACSim+ AHU model that was used in this work is available within the ASHRAE Research Project 1312 Results²⁵.

The data acquisition system that is implemented in FLEXLAB comprises a custom-built National Instruments platform utilizing distributed Compact RIOs (cRIOs), PC-based servers and workstations, and a Unix-based database running sMAP2.0²⁶. Data are typically collected and recorded at a one-second (1 Hz) rate with averaging to one minute for database storage – suitable for most research purposes. 1 Hz data are stored in CSV files for use as needed. Control and data acquisition are implemented over the same architecture, with most control sequences running in the NI TestStand environment. A simple API also allows remote data acquisition and control using text-based messages, which allows use of other programming or scripting environments such as python or java. Experimental data are accessed from sMAP using a browser-based GUI or via text-based query.

At the Flexible Research Platform, the data acquisition system utilizes Campbell Scientific data loggers. It includes measurements of the zone set point temperature and humidity, supply and return air temperature and flow rates, and energy consumption of individual components including compressor, condenser, supply fan, VAV reheating. Data are typically collected and recorded at a one-second rate with averaging thirty seconds for database storage. The data file format is CSV, with automated transfer from the data loggers to storage on an ORNL internal server, at time resolutions of 30 seconds, 1 min, 15 min, and 60 min intervals.

Data Records

The data are stored on figshare and on OpenEI, a wiki-based platform that supports public sharing of, and access to data and analyses related to renewable energy and energy efficiency. Summarized in Table 9, the dataset¹² comprises a collection of six comma separated value (CSV) files. Each CSV file represents a single combination of system configuration and experimental or simulated data creation approach. The data are minute-frequency time series measurements of the system operational parameters that are most commonly available to FDD algorithms in typical commercial buildings. Time stamps are in the first column of each file, and presented in the format *m/d/yy h:mm*. The final column of each file contains a binary indicator of the ground truth information on whether or not a fault is present.

Data file	System	Data provenance	Total file size
SZVAV	Air handling unit: single zone variable air volume	Experimental	1.2 MB
SZCAV	Air handling unit: single zone constant air volume	Experimental	1.6 MB
MZ-VAV-2-1	Air handling unit: multi-zone variable air volume	Experimental	1.8 MB
MZ-VAV-2-2	Air handling unit: multi-zone variable air volume	Simulation	2.9 MB
MZ-VAV-1	Air handling unit: multi-zone variable air volume	Simulation	22.7 MB
RTU	Rooftop unit	Experimental	9.9 MB

Table 9. Files and size of each file in the full dataset, as well as system of focus and provenance.

The set of CSV files is accompanied with a data ‘inventory’ file that describes:

key information necessary to understand the content and scope of each data set, including:

- An overview of the data set, who created it, and whether it was generated through simulation or physical experimentation
- Building and system information
 - Model or experimental facility description
 - System type and diagram
 - Control sequences
- Data points
- Input scenarios for faulted and fault-free conditions represented in the data
 - Fault types
 - Fault intensities
 - Method of fault imposition
 - Fault occurred time

Technical Validation

The technical quality of the dataset can be understood through 3 primary lenses: 1) accuracy of measurement in the facilities; 2) accuracy of the simulation models; 3) accuracy of the ground truth information on the presence or absence of faults and their severity. Description and illustrations of each are provided in the following.

Facilities measurement

The facility measurements that are included in the dataset comprise sensor data as well as data that indicate equipment status and control commands. Sensor data span temperature, relative humidity, power, pressure, and air flow. Equipment status and control commands encompass parameters such as valve and damper control commands, temperature setpoints, and operational modes.

Specifications of the relevant FRP and FLEXLAB sensors are provided in Table 10. Many more sensors are available in the facilities, however we report only those used in the dataset described in this paper).

Facility	Sensor	Measurement	Accuracy
FLEXLAB	BAPI BA/10K-2(XP)-D-12"-BB thermistor	Temperature	$\pm 0.1^{\circ}\text{C}$
FRP	Campbell Sci HC2S3-L	Temperature, relative humidity (RH)	$\pm 0.1^{\circ}\text{C}$ and $\pm 0.8\%$ RH @ 23°C
	Continental Controls WNB-3D-240P	Power	$\pm 0.5\%$ of reading
	Omega PX409-750-A5V pressure transducer	Pressure	$\pm 0.08\%$ best straight line maximum
	Air Monitor fan evaluator paired with Veltron DPT2500-Plus transmitter	Air flow	DPT2500, 0.25% of natural span, including hysteresis, deadband, nonlinearity, and nonrepeatability Fan evaluator $\pm 2\%$

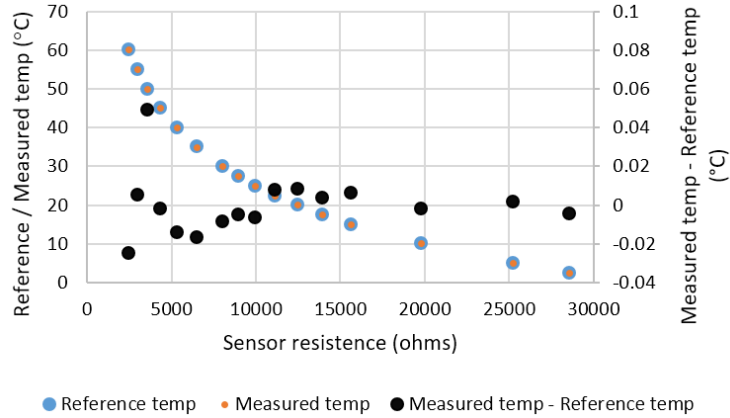
Table 10. Specifications of relevant sensors from the FLEXLAB and FRP experimental facilities.

Facility sensors are periodically calibrated using a variety of standard approaches based on ISO/IEC 17025²⁷, which specifies requirements for testing and calibration laboratories. To illustrate, a detailed example is provided for the water-bath calibration of the FLEXLAB AHU temperature sensors (thermistors).

The supply and return air temperature sensors are inserted in the appropriate duct, and the mixed air temperature sensor is placed in the AHU mixed air plenum, which is open to return air and outside air (Figure 1). The calibration is conducted using a 0.01°C accuracy reference sensor (US sensor USP3021), an Agilent/Keysight 34970A data acquisition unit, and an Anova water bath. All Agilent measurements are conducted to ensure that the measurement of the testing sensor is consistent with the reference sensor, and they use temperature step sizes of 5°C to bound the range of interest.

An example of a supply air temperature calibration result from this process is shown in Figure 5, showing calibration of experimental sensors to within 0.02°C of the reference sensor. The normal range of interest for the temperature sensors at FLEXLAB is 5°C to 45°C , so the supply air temperature sensor is calibrated in a range of 5°C to 60°C . The blue dots and orange dots are the temperature measurements from the reference sensor and the newly calibrated supply air temperature sensor respectively. The black dots which are plotted on the secondary y-axis represent the difference between the newly calibrated sensor to the reference sensor. The plot shows that across the full range of interest the measurements of the newly calibrated sensor (orange dots) match those of the reference sensor (blue dots) to within $\pm 0.02^{\circ}\text{C}$. There is one outlier (0.06°C) at the low-resistance, high-temperature (left) side of the plot. Temperatures below about 35°C are very closely aligned with the reference sensor, with offsets within $\pm 0.01^{\circ}\text{C}$.

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Figure 5: AHU supply air temperature water-bath calibration.

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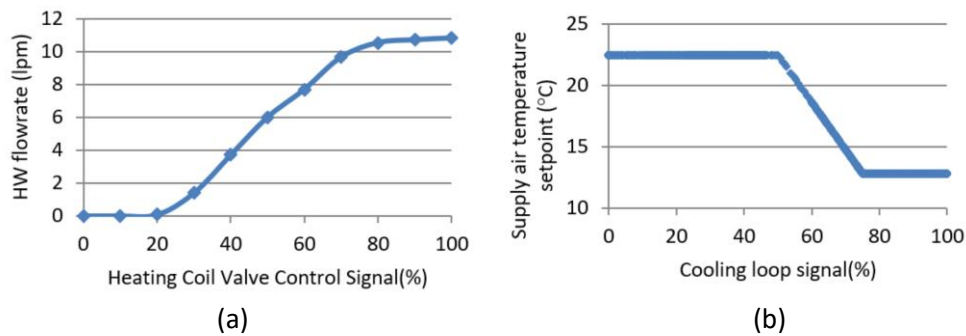
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To verify the data for equipment status and control commands, the results from functional tests were leveraged. The primary purpose of functional tests is to ensure that system operation is consistent with the designed control sequences, and reflective of fault-free operational behavior. A failed functional test would indicate incorrect implementation of control logic, equipment faults, or inaccurate system-reporting of status and command data.

For example, Figure 6(a) shows the characteristic curve for the FLEXLAB AHU heating coil valve, heating water flowrate versus heating coil valve control signal. As expected, once the valve is opened past 20%, the flowrate steadily increases until reaching its maximum value at the 80-100% open position. Similarly, Figure 6(b) shows supply air temperature setpoint versus cooling loop control signal. It indicates that the cooling loop control signal modulates between higher and lower values to drive a supply air temperature setpoint reset in accordance with the defined control sequence.



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Figure 6. Examples of functional test results from FLEXLAB.

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Figure 7 shows a functional test from the FRP RTU, in which the control command for RTU fan speed (%) signal is observed to appropriately track with measured RTU fan airflow (L/S), as the signal is increased from approximately 40% to 100%.

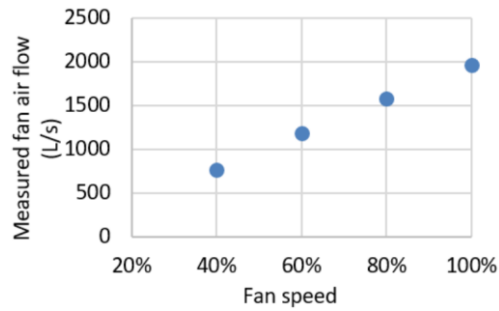


Figure 7. Examples of functional test results from the FRP.

Simulation models

An EnergyPlus-Modelica model that was used to generate the simulated data for multi-zone AHU #1²⁸. The US Department of Energy's large office Commercial Reference Building Model²⁹ (hereafter referred to as *Reference Building*) in EnergyPlus was used to calculate the building thermal load, as the *Reference Buildings* are taken to represent reasonably realistic building characteristics and construction practices and are widely used in the evaluation of building design and operation³⁰. Since control processes are idealized in EnergyPlus, and may not capture short-term behaviors of HVAC systems, the HVAC systems in the large office *Reference Building* were re-implemented with components from the Modelica Buildings Library to model the dynamic behaviors. To retain realistic system response in the Modelica representation, the system performance curves from the *Reference Building* were transferred to the Modelica representation. Modelica component and system models have been validated using empirical validation, comparative testing and analytical verification. Most relevant to the models used in this work, comparative testing, in which results are compared with other simulators, has been used for the multi-zone airflow models³¹. In addition, analytical verification, in which results are compared with exact solutions, has been used to validate most of the individual component models in Modelica, such as for heat and mass transfer and storage, for flow resistance elements such as valves and pipes, for fan models and for radiosity transport models³².

The HVACSim+ model that was used to generate the simulated dataset for multi-zone VAV AHU #2 was validated with the experimental data from the physical unit that the model was created to represent. This validation is detailed in²⁵. To validate the model under fault-free conditions, steady state experiments and dynamic experiments were conducted to generate the experimental data for the comparison. Energy indices (i.e. electrical energy consumed by return and supply fan, heating water energy consumed by heating coil, and chilled water energy consumed by cooling coil) were used to compare the simulated energy consumption with real energy consumption during the experiment. Operational indices (i.e. temperature, air flow, and control indexes) were used to compare the simulated operational variables with the actual measurements. To illustrate, the temperature operational indices that were used for model validation are provided in Table 11. In the Table "max diff." indicates the maximum difference of the hourly average values between simulated values and experimental data. Maximum differences less than 20% of the typical value were taken as sufficient.

Indices			Summer in 2007			Winter in 2008		Spring in 2008		
			8/19	8/25	9/4	2/16	2/17	5/2	5/3	5/9
Temp °C	Supply air	typical value	12.7	12.7	12.7	18.3	18.3	12.7	12.7	12.7
		max. diff.*	-0.9	-0.7	-0.5	-0.2	-0.05	-0.4	-0.8	-0.4

	Mixed air	typical value	23.8	23.3	23.3	12.7	13.3	18.3	13.3	8.3
		max. diff.	-0.4	-0.7	-0.8	-0.7	-0.2	-2.3	-1.0	-1.3

Table 11. Example of temperature operational indices used to validate a simulation model with experimental data.

To validate the model for faulted conditions, the simulated operational data under a specific fault were verified to ensure that they reproduced the major fault signature obtained from the facility measurements. An example is shown in Figure 8. The plot shows measured and simulated the heating coil outlet air temperature under faulted and un-faulted operation. The simulated temperature (brown solid line) during faulted operation is much higher than the simulated temperature during fault free operation (light blue dashed line). This offset behavior was consistent with that observed in the physical measurements, shown in red (faulted operation), and dark blue dashed (un-faulted operation).

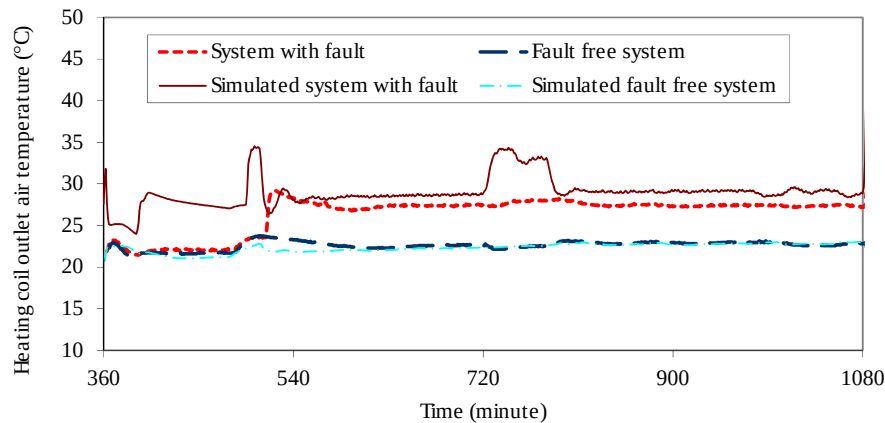


Figure 8. Heating coil outlet air temperature associated with the presence and absence of a leaking heating coil fault.

Fault ground truth

Ground truth assignment of the data as faulted or fault-free, was validated first through functional testing (experimental data), and engineering logic (experimental and simulated data). Described in the preceding section on *facility measurement*, functional tests ensure that system operation is consistent with the designed control sequences. Whereas a failed functional test can indicate faulted equipment (including sensors and status/command data), or incorrect implementation of control sequences, a successful functional test can verify fault-free operation. For example, Figure 6(a) suggests the heating coil valve moves smoothly without the presence of any pre-existing stuck or leakage faults. Similarly, Figure 6(b) indicates that the supply air temperature setpoint was reset according the defined sequence that was in place.

After functional testing, the data for each fault-present or fault-free case were visually inspected to validate the presence or absence of faults (and their severity). Engineering logic and knowledge of the implemented control strategies was applied to confirm that the “fault free” scenario and “imposed fault” scenario were indeed reflected in the data trends. Figure 8 provides an example to validate the fault signature of the imposed leaking heating coil fault, in which the heating coil outlet air temperature is confirmed to increase, as compared to the fault-free system. Figure 9 illustrates another example, for a cooling coil, stuck at a severity of 100% (i.e., fully closed, or 0% open). SAT is supply air temperature; SAT Sp is

supply air temperature set point; CV-POSN is cooling coil valve position. The data in the figure show that when the fault was imposed, the cooling coil valve position was overridden a fixed 0% position. The expected symptom of supply air temperature much higher than the setpoint was then observed, validating that the fault was correctly imposed.

Similar checks were conducted for each of the faults in both the experimental as well as the simulated data sets.

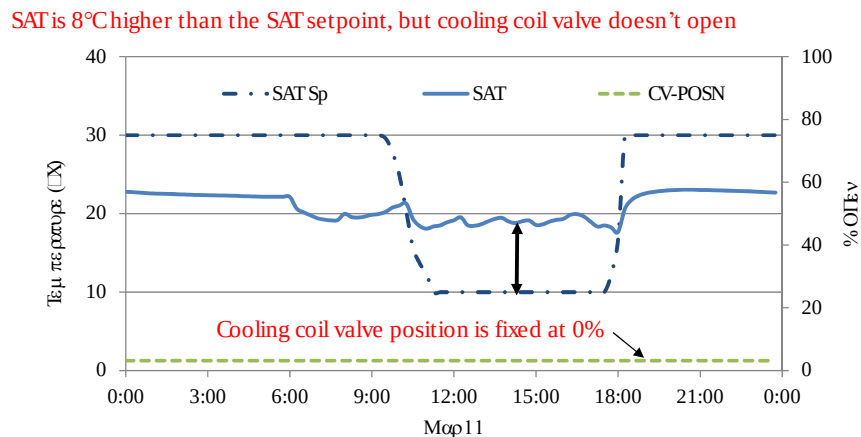


Figure 9. AHU operational data for an imposed stuck cooling coil valve fault.

Usage Notes

A complete inventory of the data was developed to support users in interpreting the content and form of the data, and the corresponding HVAC systems, controls, and faults. The data itself comprise time series that can be analyzed with whatever software tools the user elects to implement. The data are provided at 1-minute intervals, and can be resampled as needed to fit the needs of specific applications.

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Author contributions

Jessica Granderson led the overall research effort and the development and writing of the manuscript.

Guanjing Lin co-wrote the manuscript, synthesized the datasets, and led the technical validation and documentation of the data.

Ari Harding designed and implemented the methods of fault imposition for the experimental faults created using the FLEXLAB test facility.

Piljae Im designed and implemented the methods of fault imposition for the experimental faults created using the FRP test facility, and contributed to the associated documentation.

Yan Chen designed and implemented the methods of fault imposition for the simulated faults created using Modelica.

Competing interests

The authors declare no competing interests.

Packaged Scalable Energy Information Systems for Hotels

Abstract

Purpose

Building Energy Information Systems (EIS) are performance monitoring software, data acquisition hardware, and communication systems used to store, analyze, and display building energy data. Some \$60 billion is spent annually on wasted energy in U.S. buildings, and actions taken based on EIS data can enable operational energy savings of ~10 percent in the U.S. commercial sector (~2 quads of primary energy). However, EIS adoption is low due to various technical and market challenges. This paper provides technical specifications for standardized EIS packages that can help overcome barriers and accelerate scale.

Design/Methodology/Approach

A five-step process was followed:

1. Identifying business drivers as key determinants for hotel sector-specific packages
2. Addressing heterogeneity to develop standardized, tiered packages
3. Determining performance metrics for key stakeholders
4. Recommending streamlined data architecture
5. Developing visualization enabling insights and actions

Findings

Technical specifications for two tiers (entry and advanced) of EIS packages for hotels have been developed. EIS vendor, integrator, and client organization's facilities and IT staff have been considered as key stakeholders. Findings from six field demonstrations show benefits of (i) **cost-effectiveness**, through reduced transactional, first, and operational costs, (ii) **scalability**, by accommodating heterogeneity across the building sub-sector, (iii) **simplicity**, by integrating meters, gateways, and software in the package, and (iv) **actionability in organizations**, across various decision making levels.

Value

Building owners and operators can use these specifications to ease procurement and installation of EIS in their facilities. EIS software vendors can use them to develop new product offerings for underserved sectors.

Packaged Scalable Energy Information Systems for Hotels

1. Context

Although architects and engineers target energy efficiency in building design, operations of buildings differ from the design intent. Every minute, day, week, and month, in countries spanning the globe, from United States to India, large amounts of the energy consumed in buildings is not actually utilized for any purpose. For instance, one of the largest building energy loads, heating, ventilation and air conditioning (HVAC) systems often fail to meet performance expectations due to various faults, poor maintenance and controls, and improper commissioning (Roth, 2005). It is normal for building equipment to experience *operational stray* (Henderson and Waltner, 2013), where actual energy use is higher than designed. A building's actual energy use can be two to five times higher than designed (Roth et al., 2005). When the problem persists undetected over long periods of time, it can lead to an estimated 15 to 30 percent of energy wasted in commercial buildings (Katipamula and Brambley, 2005).

The business-as-usual approach of tracking energy use through monthly utility bills is typically too late (being post-facto) and too coarse (being at a whole building level) to identify causes for wastage. Optimal performance requires access to higher granularity of energy consumption data, and more timely analysis. It requires that building energy use be continuously monitored and managed to curtail operational stray, capture deeper energy and dollar savings, and attain energy performance targets.

On average, U.S. hotels spend approximately 6 percent of revenue on utilities, but historic and luxury properties may see energy costs hitting 10 percent or more (National Grid, 2004). A 10 percent reduction in energy consumption has the same financial effect as increasing the average daily room rate by \$0.62 in limited-service hotels and by \$1.35 in full-service hotels (ENERGY STAR, 2007). Trends from the hospitality industry show that utility costs are the second largest operational cost, after labor (Gaggioli, 2016). Some forward-thinking hotel owners use proprietary tools to capture monthly energy, waste, and water consumption data to identify trends, track performance, and compare the same against benchmarks, to inform decisions for efficiency actions for hotels in their portfolio. For instance, a large hotel chain reports that since implementing this tool in 2012, their hotels have saved \$185 million in avoided costs. It also states that 75 percent of frequent travelers care about sustainability and one-third of business travelers actively seek environmentally friendly hotels—an indicator that this enterprise strategy could potentially attract clients to build up topline growth (Better Buildings, 2015).

Such trends are also relevant in emerging economies, such as In India, where the hospitality sector has been growing at a cumulative annual growth rate of 14 percent (Indian Brand Equity Foundation, 2017); major international hotel chains are planning to set up or expand significant properties within the next decade. The high growth rate of this new hospitality footprint affords a substantial opportunity for energy efficient design and operations.

The good news is that information technologies offer opportunities to reduce building energy demand and associated greenhouse gas emissions. Energy Management and Information System (EMIS) is a broad family of tools and services to manage commercial building energy use. These technologies include, for example, energy information system (EIS), equipment-specific fault detection and diagnostic systems, benchmarking and utility tracking tools, automated system optimization tools, and building automation systems. (Lawrence Berkeley National Laboratory, 2015). A key energy efficiency enabling technology are energy information

systems (EIS), broadly defined as a system with performance monitoring software, data acquisition hardware, and communications used to store, analyze, and display building energy data.

Through an EIS one can access energy data to identify consumption and patterns to track energy use, manage demand charges, validate utility billing, identify waste, benchmark building energy performance, and help identify short-term and long-term goals and actions toward energy efficiency opportunities toward 20 percent savings. A cost-benefit study conducted by Lawrence Berkeley National Laboratory (LBNL) across 26 enterprises' building portfolios shows that participants achieved year-over-year median site and portfolio savings of 17 and 8 percent, respectively; they reported that this performance would not have been possible without the EIS. The median five-year cost of EIS software ownership (up-front and ongoing costs) was calculated to be \$1,800 per monitoring point (kilowatt meter points were most common) (Granderson et al., 2016). Facility owners, energy and sustainability managers, and facility operators can use the data analytics provided by an EIS to drive energy efficiency, improve building performance, and save energy costs.

Larger hotels and enterprises may be able to leverage economies of scale, and with bigger infrastructure and personnel budgets are well positioned to implement customized EIS solutions. However, small- to medium-sized hotels such as high-growth business hotels remain underserved relative to efficiency services, owing to tighter margins and lack of energy management staff. In these facilities current EIS solutions can be cost-prohibitive and present a high bar for entry, limiting adoption. Hotels are a prime example of a case where the need for guest comfort and services can cause significant energy waste, and where cost-effective, easy-to-use EIS can play a critical role in achieving energy savings.

2. R&D Motivation

Despite the strong rationale for EIS as a key emerging technology that enables building energy savings, and dramatic increases in the number of EIS market offerings, several barriers impede their broad adoption.

First, enterprise decision makers typically have insufficient awareness of the energy cost and use, and the role that EIS can play in minimizing those costs. Executives can be unaware that 3 to 5 percent of their revenue is being spent on energy, or even if they do, they perceive that to be a fixed, unmanageable cost (Forbes, 2015). With such little awareness of the problem, the adoption of a solution is a challenge. Furthermore, even if owners do realize the value that an EIS can deliver, they still face a fragmented market of building energy management systems, tools, and vendors that makes it challenging to appreciate, procure and specify a relevant EIS for their building or enterprise. Enterprises are also concerned about data security.

Second, even within the hotels sector, buildings are heterogeneous. Energy Information Systems are generally customized on building-by-building basis, which is typically a time-consuming process that requires a high level of skill to configure the EIS, a robust metering infrastructure, drawing insights from data analyses, and provision of services. This customization drives up the transaction cost for an EIS.

Third, currently available EIS tend to be too complex to serve the basic needs of small- to medium-sized hotels. The operations staff are also generally not trained in the installation and use of typically sophisticated EIS and need to rely on vendor services, which creates additional ongoing costs for the organization.

From a vendor perspective, it is difficult to market and sell EIS solutions for small- to medium-sized facilities. The foremost challenge is high upfront technology cost, which is a function of the high transaction cost required to customize EIS on a building-by-building basis. If coupled with ineffective use of sophisticated EIS technologies, or high vendor cost for services, return on investment (ROI) may not prove justifiable. Factors contributing to the wide range of upfront technology costs include the number of monitored points (e.g.,

only whole building energy usage vs. extensive sub-metering), extent of software features, and configuration needs.

These current practices of building-wise customization and high transaction costs have led to low adoption of this innovative emerging technology in all but the largest facilities and most sophisticated campuses and enterprises. The adoption of EIS is largely limited to large organizations or buildings with large energy expenditures that justify the first and ongoing high cost of the EIS.

The motivation of this work is to overcome these challenges through design and engineering of cost-effective packaged EIS solutions that can help drive adoption, especially in underserved building sectors, in order to help curtail energy waste and enable optimal use of operational energy across the building stock. For this study, we focused on business hotels—the largest group of hotel types that primarily cater to business travelers. These are typically underserved, but with significant opportunity for energy efficient operations. We anticipate the findings are extensible to small- to medium-sized convention centers, airport hotels, and budget to mid-range service hotels.

3. Concept of a Packaged EIS

We propose simplified, low-cost EIS packages—“EIS-in-a-box”—that overcome barriers such as installation and commissioning complexity, difficulty of use, and high cost. We have developed packaged EIS broadly applicable across the hospitality sector, rather than needing high customization at a building-by-building level. We derived the technical requirements for EIS packages with three predefined components, as shown in Figure 1, namely:

- (i) Data acquisition and communication through sub-meters, and gateway connectivity
- (ii) Back-end software for data access, storage, and analysis
- (iii) Front-end visualization and user notifications

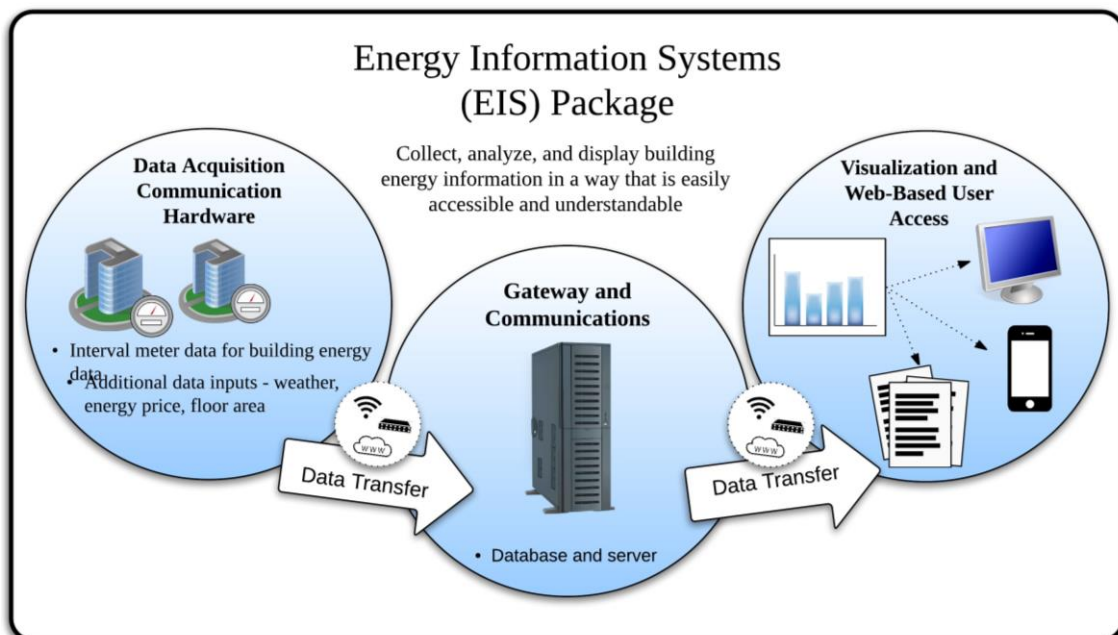


Figure 1: The three components of an Energy Information System (EIS)

Our proposed simplified, sector-specific EIS packages are detailed in subsequent sections. These packages are intended to obviate the need for customization, and enhance ease of procurement, installation and use, thereby enabling scale-up and wider adoption in the building sector.

4. Key Development Considerations

We took into account two key considerations to develop packaged EIS-in-a-box for hotels:

- (i) How to **frame typical transaction costs** for the procurement, installation, and use of EIS
- (ii) How to characterize **the heterogeneity** in the hospitality sector, and accommodate its influence on package hardware design and engineering

These considerations are detailed below.

4.1. Transaction cost framework

Transaction cost is defined as *costs other than the money price that is incurred in trading goods or services...and activities (that) involve opportunity costs in terms of time, effort and money* (Johnson, 2005).

The transaction process involves the time and effort required to deliver a product and service. For an EIS solution, typically there is a four-step transaction process (as shown in Figure 2) that can easily take several months or more. Each step requires time and effort that leads to overhead costs borne by the vendor that are typically passed on to the client. Through a series of structured interviews with seven EIS vendors, we determined that a significant portion of the cost is in implementing steps 1 through 3 (i.e., client recruitment, system configuration, and system integration). Our analysis showed that there exist hidden technical opportunities for reduction in transaction cost during each of these steps. For example, the time taken for steps 2 and 3 may be shortened if the electrical circuitry is viable for simple sub-metering. Additionally, metering and installation costs can be reduced significantly if a prioritization is done to select core sub-metering points based on an understanding of typically high and controllable loads. Similarly, offering value across the stakeholder ecosystem through a simple, secure system with reports and metrics targeted across facilities, Information Technologies (IT) staff, and executives can ease the buy-in process and time required for integration.

Our intent has been to comprehensively discover such opportunities and develop relevant technical requirements for packaged EIS that allow the steps to be streamlined to reduce the time, as well as first and recurring costs, for installing and operating an EIS. Our goal is that within a compressed period of transaction time (on the order of a few days) the product requirement can be fulfilled with reduced first cost (~30 percent less) for both vendors and users. Additionally, by using our user interface guidelines, in-house facility managers can be better equipped to understand the energy behavior of their facilities, conduct the first set of actions for troubleshooting, and easily operate and maintain the system, all leading to reduced operating and service costs. Beyond that, deeper vendor expertise can still be sought as needed for energy projects and upgrades. By offering an enhanced value proposition for an EIS, it is our intent that vendors and integrators experience fewer barriers to increased market penetration using existing hardware, communications devices, and distribution channels. That, in turn, should lead to better sales in higher volume, which may further lower the price point for a packaged EIS-in-a-box.

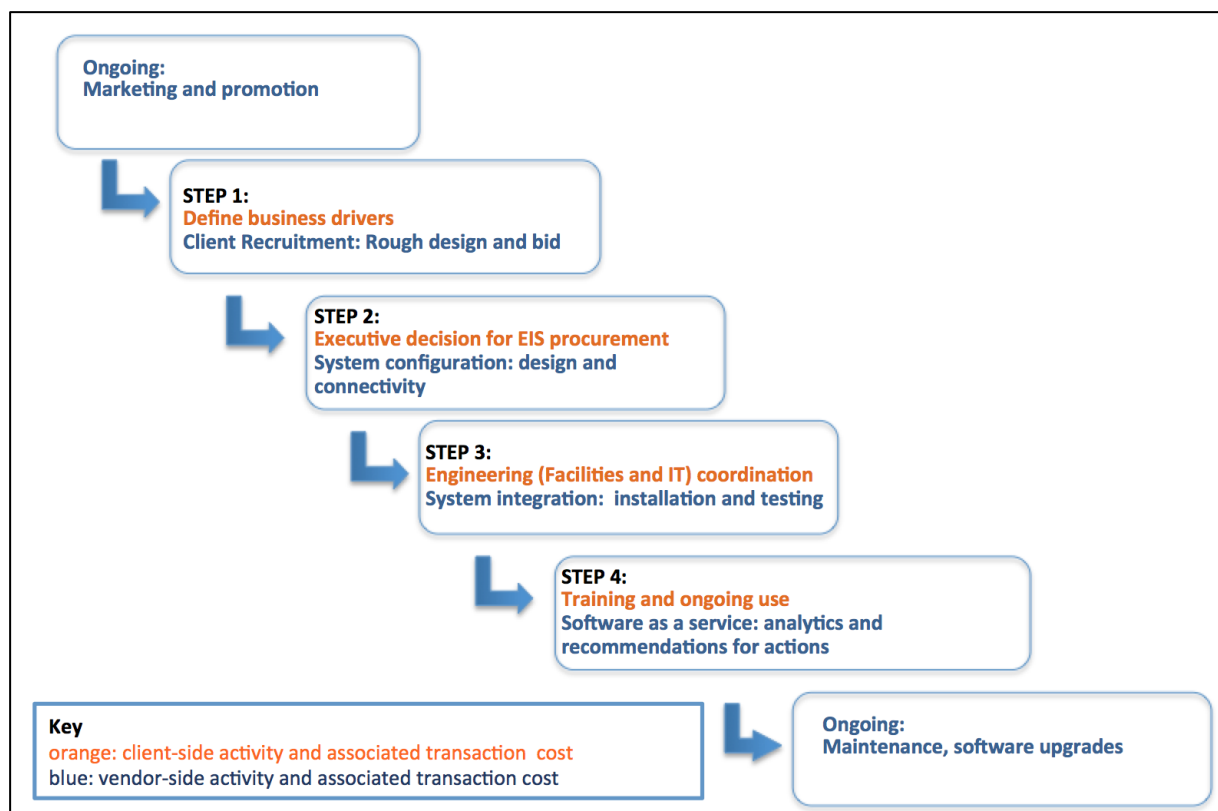


Figure 2: Typical transaction cost framework for the specification, installation, and use of an energy information system (EIS)

4.2 Heterogeneity in the hotels building sector

A second key consideration is the extent to which the energy and metering infrastructure and management approach varies from hotel to hotel. This heterogeneity affects the extent to which a packaged standardized solution can be broadly applicable. We investigated the types of hotels to understand the physical and organizational attributes that drive energy use to enable a relevant package design.

Typically, 75 percent of a hotel or motel's total energy use can be attributed to space heating, lighting, and cooling combined, with cooling and lighting alone comprising half of the building's energy consumption, as shown in Figure 3. However, there is a broad range of energy use intensity across lodging facilities, ranging from 15,000 Btu per square foot to over 300,000 Btu per square foot (U.S. Department of Energy, 2016).

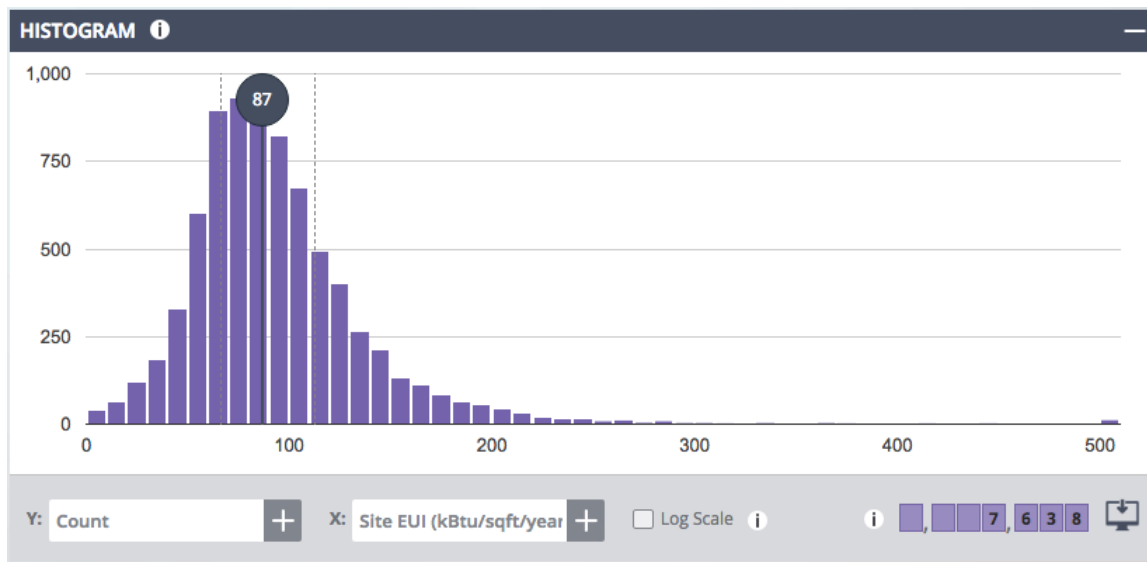


Figure 3: Screen shot from the Building Performance Database (BPD) for 7,638 U.S. hotels. Median Site EUI is 87 kBtu/sf. 25th, 75th percentiles are 66 and 113 kBtu/sf, respectively. (Source: U.S. DOE, 2016)

4.2.1 Mapping physical infrastructure

There is wide diversity in hotel facilities, classified by parameters such as size of the hotel, target markets, level of service, and ownership, as shown in Figure 4. These parameters are significant determinants for the physical design and operations of hotel facilities, and thereby a key consideration to assess how and where an EIS may help impact energy use. Another important consideration is to analyze and understand potential saving opportunities and EIS design in new construction, as compared to those possible in existing facilities.

Classification of Hotels				
Physical attributes			Organizational attributes	
Classification by size	Classification by target market	Classification by location	Classification by levels of service	Classification by ownership and affiliations
- Small hotels: up to 25 rooms - Mid-sized hotel: 26-99 rooms - Large hotels over 100 rooms	- Business - Suites - Transit - Extended Stay - Services apartments, - Resorts - Bed and Breakfast - Heritage - Boutique - Casino hotels - Convention centers	- City/Downtown - Suburban - Airport - Highway motels - Floatels	- World class - Mid-range - Budget	- Independent hotels - Chain hotels - Time share - Condominium hotels

Figure 4: Heterogeneity in the hotels building sector, based on physical and organizational attributes

4.2.2 Organizational characteristics and business drivers

Organizational considerations in developing the packaged EIS included factors such as the value attributed to energy, and staff motivation and skill available to take energy-related decisions and actions. Even within an organization, a diversity of stakeholders have influence, ranging from facilities staff operating on a day-to-day level, and IT staff who take decisions regarding the information technology infrastructure, to facilities executives and sustainability managers who may make investment and high-level decisions. While facilities staff may consider building automation systems important for primarily building operations and management, the value of energy efficiency is often far from being fully appreciated or visible, or perceived as being business-relevant. Understanding and being responsive to organizational business drivers and needs is a key precursor for an effective EIS.

These considerations were taken into account during the design of the packaged EIS and are detailed in the next section.

5. Approach to Determine Specifications for Packaged EIS

We followed a five-step process to determine the specifications:

- Step 1: Determining business drivers for the building sector as the core determinants for the package
- Step 2: Addressing heterogeneity of facilities through a tiered package structure
- Step 3: Identifying metrics for key stakeholders and decision makers for which data would be derived
- Step 4: Recommending a “picklist of loads” for controlling data cadence
- Step 5: Forming insights and actions based on data analysis presented in simplified user interfaces

Each step is detailed in the sub-sections that follow.

5. 1 Determining business drivers for hotels

Through a literature review of trends related to facility energy use in hotels, and interviews with industry collaborators, we identified five business drivers for hospitality organizations. These drivers, and their impact on the design of an EIS package is detailed below:

- (i) **Monitor energy performance of their facility.** To address this driver, an EIS should provide near real-time time-series information on facility energy use and quantify changes in energy use over time.
- (ii) **Track cost and demand** to understand the financial implications of energy use, and wastage; identify base and peak demand, and assess system size and efficiency of mechanical equipment. An EIS should support the establishment and monitoring of utility budgets and costs, and develop annual energy reports.
- (iii) **Benchmark energy performance** to have an effective yardstick for demand, efficiency, and energy use targets by comparing the facility’s energy performance against a peer. Hence an EIS should perform “cross-sectional benchmarking.” Second, it should benchmark the facility against itself, (i.e., validate the energy performance against the design intent, initial commissioned operations, or base period of performance). Hence, an EIS should perform “longitudinal benchmarking” (Granderson et al., 2013).

- (iv) **Identify and track energy efficiency projects.** An EIS should provide information to identify, understand, and mitigate risks of undertaking energy efficiency measures, track persistence in savings through any implemented projects, and track improvements over time.
- (v) **Be environmentally sustainable and compliant** to track greenhouse gas emissions, for instance for benchmarking or carbon disclosure program reporting and city energy disclosure ordinances. An EIS may provide this option.

Once we identified the business drivers for hotel organizations, we mapped features and functionality of packaged EIS to address these drivers. There are operational or cost metrics associated with these drivers. Our goal for EIS-in-a-box packages is to provide information relevant to operational performance metrics pertinent for the facilities staff, as well as cost metrics that tie back to the topline revenue and profit, pertinent for executive-level decision-making. The EIS should acquire energy data from consistent sub-metering points to feed into the analyses for performance and cost metrics. Further, the EIS should provide effective visualization to make the energy information accessible, optimum, reportable, and actionable. This logic flow for the design of packaged EIS is shown in Figure 5.

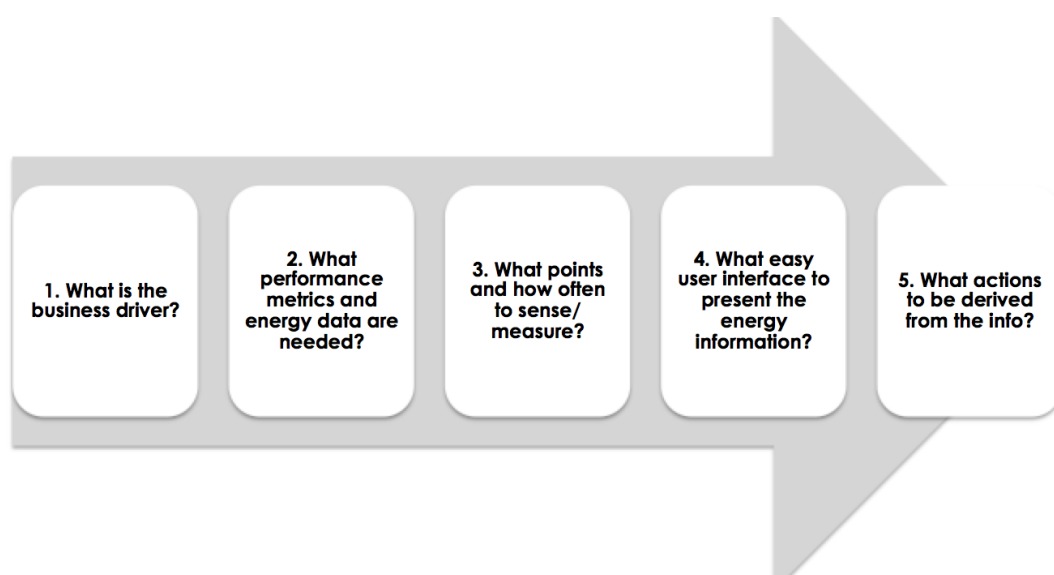


Figure 5: Logic flow for developing a packaged EIS-in-a box based on prioritized business drivers for a building typology

5.2 Addressing heterogeneity through a tiered package structure

As mentioned earlier, hotel facilities and organizations are heterogeneous, and their staff possess varying levels of skills and motivation to save energy. To address this heterogeneity, we developed two distinct tiers of EIS packages: Entry (Tier 1) and Advanced (Tier 2). These tiers represent a light-touch and medium-touch approach respectively, to provide technology options that are commensurate with organizational skills and motivation. The difference between the two tiers lay in their objectives, target user/audience, and features and functionality—relative to the Entry tier, the Advanced tier marks an increase in complexity, cost, and energy savings potential.

The Entry Tier 1 EIS package for hotels offers organizations that traditionally have little or no visibility into their building energy footprint, and whose building managers who have extremely limited time and resources, to obtain only the most important information. The Entry Tier 1 package is a “foot in

the door” to familiarize users with installation, use, and benefits of a simple EIS. In such organizations, a monthly utility bill may be the only currently available energy use information, which is post-facto.

On the other hand, organizations that recognize the value of energy and provide some resources for their building managers to monitor their building’s energy usage may be better served through an Advanced Tier 2 solution. This is targeted toward hotel owners and managers who have a higher awareness and interest in the benefits of energy efficiency and carbon accounting, and the ability to spend investment dollars and staff resources commensurately.

Figure 6 shows, for each key business driver, how the Tier 1 and Tier 2 packaged EIS solutions compare with the custom EIS offerings common in today’s market. We note that users could begin with Tier 1 and evolve its capabilities to a Tier 2 level over time, as their needs change from “introduction to quick energy wins” to “best practices for ongoing maintenance of energy performance.”

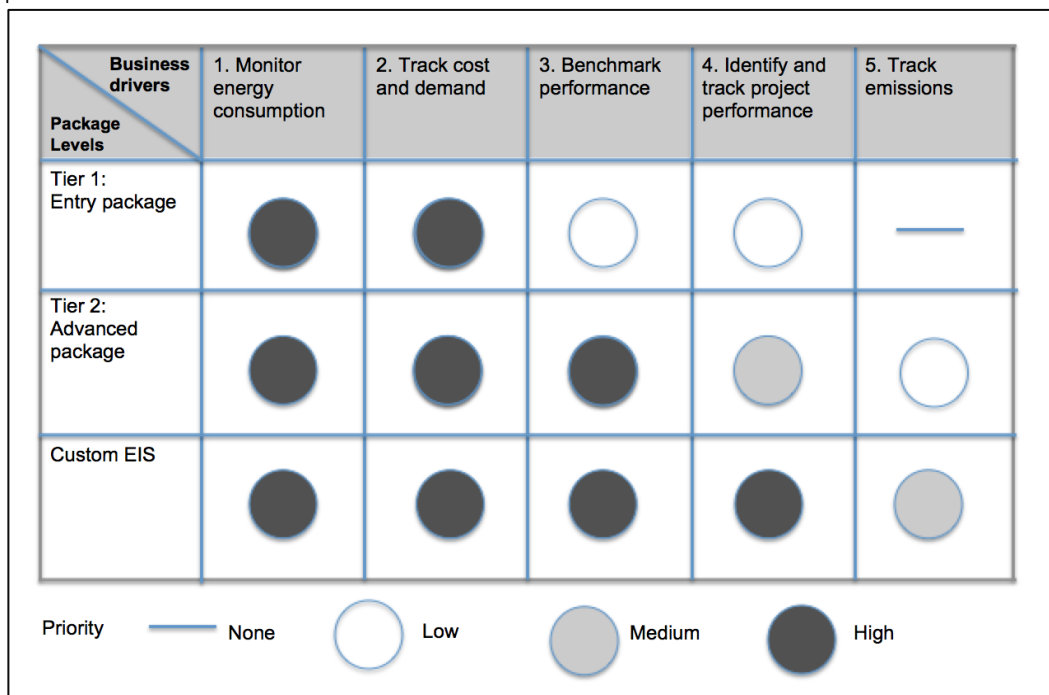


Figure 6: Entry and Advanced tier functionality, built around prioritized support provided to business drivers

5.3 Identifying metrics for key stakeholders and decision makers

In the spirit of making energy information visible and valuable across the hotels’ stakeholder ecosystem, we identified two primary types of users for the packaged solution: (1) facility managers who track energy granularly at a daily or sub-daily basis, and (2) management/executives who would view energy and its financial implications quarterly or annually (at a minimum). Further, we identified the two types of audiences’ critical questions and metrics of interest, to track and report on a rapid, short-term basis (daily/weekly) (Figure 7) and a long-term (quarterly/annual) basis (Figure 8). We generated recommended daily and annual dashboards with easy-to-use charts. Our recommended dashboards targeting these two audiences are shown in Section 6: Results.

Building Pulse at a Glance: Facilities Dashboard with Five Metrics

Primary Audience: facility managers, engineering staff

Timescale: daily, weekly

1. What is my hotel's whole-building **Absolute Energy Consumption**?
 - kWh or kBtu (or therm) per day or per week
2. What is the **normalized Energy Use Intensity** of my hotel facility?
 - kWh or kBtu (or therm) per unit square area
 - kWh/room or per occupant*
3. How is my building performing compared to past performance, i.e., **longitudinal benchmarking**?
 - kWh or kBtu (or therm) use for given day or week versus a previous time period
4. What is the **load demand per end use** of my building, and are the end-uses operating efficiently?
 - kW or kBtu/hour per time period
 - % Portion of the total energy use*
5. What is the **fuel consumption and cost**?
 - kBtu/fuel per time period
 - \$ per time period

*Tier 2 only, using building occupancy inputs

Figure 7: A list of questions and metrics that inform the facilities daily/weekly dashboard

High-level Picture: Executive Dashboard with Five Metrics, with Additional Facilities Charts

Primary Audience: executives, facilities managers

Timescale: monthly, quarterly, or annually

1. What is the **fuel consumption and cost**?
 - kBtu/fuel per time period
 - cost per time period
2. What are the trends for my facility's energy costs?
 - cost per time period
3. What is my building's **energy performance**, and how is it performing over time compared to baseline, i.e., **longitudinal benchmarking**?
 - kWh or Btu per month, quarter, or year
 - kWh or kBtu/unit square area
 - kWh/room or per occupied room*
4. How is my building performing compared to other similar facilities in my portfolio, or benchmarks, i.e., **cross-sectional benchmarking**?
 - kBtu/fuel per time period, or per unit square area, or per occupant*
 - cost per time period
5. What is **the carbon footprint** of my facility?
 - Metrics tons or pounds of carbon dioxide (CO₂) per year

Additionally, for facilities staff

6. What is the **load demand per end use** of my building, and are the systems operating efficiently?
 - kW or kBtu/hour per time period
 - % portion of the total energy use
 - kW/ton*
7. What does an **annual snapshot** of my facility look like? Is it performing well throughout the course of a month/year?
 - kW or kBtu/hour per time period

* Tier 2 only

Figure 8: A list of high-level questions that inform the monthly/annual dashboard, targeted primarily for decision making at the owner/executive level. Additional facilities-level charts are also provided for the facilities staff.

5.4 Determining a pick-list of loads to monitor in the EIS

A key approach for package design is simplification, i.e., paring down the extraneous hardware, removing any superfluous data collection, and streamlining the data analysis and visualization without adding any sophisticated new algorithms and techniques. The optimization of data collection is also driven not just by simplicity of use, but also by the emerging challenge of too much data, as can be encountered in custom EIS implementations. On one end of the spectrum are legacy meters, read 12 times a year to track consumption post-facto. At the other end, smart meters barrage readings at 15-minute intervals, providing 35,000 meter readings a year. This amount of data is an opportunity for data interrogation, however, it is costly data overload, and our target segment would find it virtually impossible to gain insights relevant to them, given their typical lack of data analysis experience.

Measuring every end use load and piece of equipment in a hotel building would require hundreds of metering points, which would be neither practical nor cost effective. Hence, only prioritized loads were recommended to be monitored, to provide the relevant data to feed into the pertinent metrics required for the business drivers.

Our aim was to reach an 80-20 solution, i.e., select approximately 20 percent of core measurement points in a hotel building necessary to provide 80 percent of the most critical information necessary for energy-based decision making. In order to select the core metering points to measure loads or end-uses, we developed a decision framework determined by following three selection criteria:

- (i) **Is the load large?** Answering this question helped to identify the most significant loads by size in an office facility. For example, space heating and cooling loads typically account for one-third to one-half of a hotel's energy use.
- (ii) **Is it a discretionary load/end-use?** It is important to characterize loads in a facility that can be controlled, managed, or scheduled by the facility manager (or even the occupant) versus loads that are too indispensable (e.g., Uninterruptible power supply, UPS) or are too regulated or distributed to be flexibly controlled or managed. For instance, a hotel conference room or business center that is used only at certain times may be easy to schedule. Lobby lighting, however, is not a discretionary load to schedule, since lighting would be required continuously for hotel operations. On the other hand, lighting may be *controllable*, and able to be dimmed or tuned based on operations and occupancy.
- (iii) **Is the load reasonably measurable?** Analyses must determine whether the electrical design provides an opportunity to sub-meter certain points, or even disaggregate data through subtraction or back-calculation, i.e., virtual load. While there is more variability in the wiring design in some hotel portfolios, many portfolios are required to follow the guidelines for electrical design, which allow predetermination of standardized points for energy monitoring.

Using this decision framework we assessed which electrical points scored high on all three decision criteria. An example is space-cooling loads in hotels: cooling loads typically account for a *significant contribution* to the energy use, and cooling is potentially able to be scheduled in a business hotel room based on occupancy, thus making it *discretionary*. Finally, HVAC loads are often on an independent circuit, making them *measurable* (sub-meterable). Hence, space cooling is prioritized as a core monitoring point. Another example is lighting. Even in situations where lighting is not disaggregated from plug loads at the panel level, there still is value in obtaining data from the mixed lighting-plus-plugs panel.

Using this method, we derived a picklist of loads (See figures 11 and 12 in Section 6: Results). This picklist is a set of core recommended energy-monitoring points for a hotel facility, and can be further broken down if the building is owner-occupied or leased. This picklist would be further used to inform technology users about the types of meters and gateways and the associated analysis and visualization that are essentially the technical requirements for the EIS package. By creating this predefined package of EIS components, we aim to reduce the usual transaction costs borne from developing custom EIS configurations on a facility-by-facility basis.

5.5 Forming insights and actions

Based on the questions to be answered in the short-term (Figure 7), we developed a daily/weekly dashboard, and based on the questions to be answered in the long-term (Figure 8), we developed a quarterly/annual dashboard. These dashboards enable facility staff to monitor and answer pertinent questions quickly and succinctly, conduct actions, and to send reports up the management chain to the executive level. Each package provides guidance on how to interpret energy data and relay information from each energy consumption analysis. Flexibility is built in for a suggested picklist of loads, incremental configuration of charts, and associated notifications based on whether it is pertinent to an Entry or Advanced package.

Common reasons for initiating energy-related upgrades in hotels and motels include customer complaints, corporate sustainability policies, frequent equipment malfunctions and shortened equipment lifetime due to years of deferred maintenance, piecemeal additions to buildings, and internal changes to existing spaces that have not been accompanied by corresponding changes to heating and cooling systems. Other reasons are previous attempts to reduce energy use by inappropriate measures, such as covering vents, and major pieces of capital equipment or building elements, such as a boiler or a roof, that are nearing the end of their useful life (ENERGY STAR, 2015). These are usually cost-intensive, one-time upgrades. What we propose through the ongoing use of an EIS is a staged approach that can reveal no-cost or low-cost opportunities that are mostly incremental, and enable persistent savings on energy costs through load reduction strategies.

Facility operators can follow a staged approach and assess the impact of each upgrade through EIS energy use tracking. This staged approach is listed below, and is shown in Figure 9:

- (i) **Schedule loads:** This involves turning off and on loads based on predetermined schedules, such as tracking and ensuring that HVAC settings in lobbies, offices, and other such peripheral and back rooms are at minimum settings during hours of low use. This strategy helps to keep the peak-to-baseload ratio high, so that off-hours setbacks are aggressively maintained for energy savings during those hours. Scheduling can be effective if there is pre-known information regarding occupancy, events, or any evening, nighttime, off-peak, or seasonal variations.
- (ii) **Control loads:** The basis of ongoing controls may be occupancy, outdoor air temperature, grid pricing, and other such measures. Examples include: using demand control ventilation for areas with variable occupancy, using economizers for free cooling when the weather is appropriate, and setting laundry hot water to 120° Fahrenheit—an optimal temperature for all hot water uses outside of the kitchen—where codes are specific about water temperature. For hallways, a simple recommended strategy is to use a combination of scheduled lighting and dimming plus occupancy-sensor controls after hours. Another example is to reduce the amount of overnight lighting used in the guest hallways by dimming controls. The results are

not only energy savings but also the unforeseen benefit of fewer noise complaints from other guests, since hallway conversations linger for a shorter time (ENERGY STAR, 2016).

Beyond these in-house actions around scheduling and controlling loads, there may be additional actions required that warrant external contractor or vendor support, such as the following:

- (iii) **Repair equipment:** If, despite scheduling and controlling loads, the equipment shows energy waste, this may indicate the need for component, equipment, or system repair.
- (iv) **Audit building/systems:** Beyond all the previous approaches, if the whole building or a system shows energy waste, it calls for executive decision making around deploying an energy audit for a deeper dive to investigate causes and further determine potential energy efficiency projects.
- (v) **Upgrade systems, equipment, and appliances:** This would be a last resort, to deploy capital resources into a physical system upgrade. The EIS will track energy costs before and after the energy efficiency projects, such as replacement with energy efficient, ENERGY STAR-labeled appliances, replacement with high-intensity fluorescent (HIF) lighting for parking lots and outdoor applications, or HVAC upgrades.

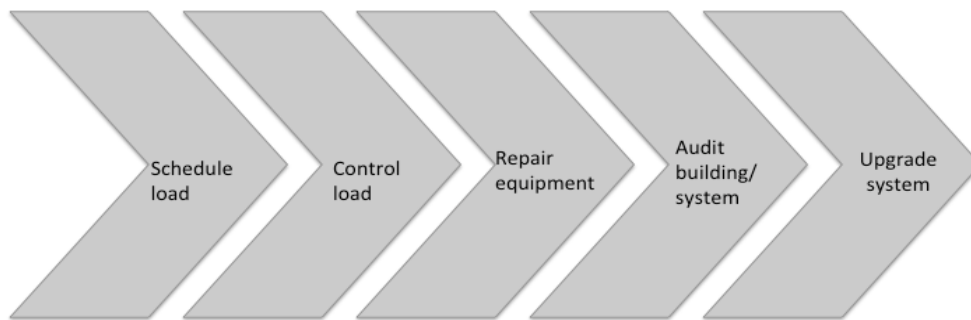


Figure 9: Staged approach to energy savings based on EIS analytics

Additional practices such as hotel reservation best practices can be followed aggressively; these can include booking rooms in clusters and renting last the cooling- or heating-intensive areas, such as rooms on top floors, at building corners, and facing west (in summer) or north (in winter). Best housekeeping practices such as covering heated swimming pools, turning off all lights, closing drapes, and setting temperatures to minimum levels in unoccupied rooms will save energy, and that will be reflected in the EIS's energy use tracking.

6. Results

The packaged EIS solutions have three components: energy meters, communications gateways, and software with a user interface. The hardware components, such as sub-meters and gateway, are specified for each tier, and are selected from off-the-shelf products that comply with the specifications, as shown in Figure 10. An important first-order supply-side interval metering is recommended for whole building electric, natural gas, other fuel and standby power. This should be followed by sub-metering of selected points and locations that are identified through the "picklist of loads" as detailed in Section 5.4 above. The whole building and systems/end-use sub-metering information acquired feeds into metrics-oriented visualization, and can help trigger action toward delving further into a certain fuel or end-use. Thus, software analysis and visualization is defined for each tier.

Section 6.1 and 6.2 describe the Entry Tier 1 and Advanced Tier 2 EIS packages.

Meters	Sub-metering Points	Physical Location	Communication Gateway	Measured Parameters	Accuracy and Turndown (U.S. and India)	Additional Inputs
Tier 1: Electric sub-meters*	Whole Building, 2–3 major loads (spaces or end uses) such as chiller plant, fan energy	1 Main distribution board (DB)	Wired between sub-meter and gateway, Wi-Fi between gateway (1) and remote database RS-485 (Modbus and BACnet) output standard for India; TCP/IP for U.S.	kWh, V, A	Class 1 according to IS13779 (India standard); 1% with 10:1 turndown (U.S. requirement)	- Bldg./space areas - Fuel supply cost
Tier 2: Electric sub-meters*	Whole Building, 7–10 major loads (spaces or end uses) such as: - chiller plant - fan energy - emergency equipment/plugs/lights	1 Main DB + Representative Spaces / Floor DB	Wired between sub-meter and gateway, Wi-Fi between gateway (1) and remote database RS-485 (Modbus and BACnet) output standard for India; TCP/IP for U.S.	kWh, kW, V, A, Power Factor, For WB: current and voltage harmonics	Class 1 according to IS13779 (India standard); 1% with 10:1 turndown (U.S. requirement)	Tier 2 - Bldg./space areas - Fuel supply cost - Operating schedules - Outdoor air temperature (OAT) from weather data
Tiers 1 and 2: Gas sub-meters Primarily U.S. package	Whole Building gas; 1 major space heating load (boiler or furnace)	1 main piping location, at all boilers/furnaces	Pulse output counting using a twister pair to gateway (e.g., pulse counting and convert to therms)	Sub-meter reads out in cubic ft, data required as therms	U.S. ANSI B109 standard; 1% with 100:1 turndown	
Tier 2: Btu sub-meter	Water cooling and heating	At chiller and boiler plant	Scaled pulse or RS-485 (Modbus and BACnet) output standard	Btu/h	Precision matched temp. Sensors, 2% accuracy 10:1 or 4% accuracy with 100:1; Standard EN 1434	

Figure 10: Metering and gateway requirements for EIS packages

6.1 EIS Tier 1, Entry package

Audience: The EIS Tier 1 Entry package (Figure 11) is targeted toward hotel owners and managers who have an interest in understanding their buildings' energy utilization, but have low skill and time. An entry-level "foot in the door" EIS package can familiarize a user with the installation, use, and benefits of a simple EIS based on information about when and how much total electricity is being consumed and wasted, primarily at the whole building level. This is a solution that most optimizes sub-meters and dashboards that can identify low-hanging fruit for energy savings. It would have an arguably lower energy savings potential, on the order of 3 to 5 percent at a whole building level. The EIS Tier 1 Entry package provides an introduction to the organization on quick energy wins.

Data Requirements: Interval meters at whole building and sub-meter levels for 2 to 3 critical points from the picklist of loads, selected on the basis of significance and controllability of the loads and ease of metering it. These loads can be at the end-use level (e.g., HVAC and all lights and plugs), for critical equipment (e.g., chiller plant). No additional user-supplied information is required.

Visualization and Functionality: Two to five preconfigured charts in visualization dashboards, as shown in Figures 13 and 14, to provide information for the features provided below:

- Simple tracking of energy consumption (KWh) to identify ongoing use and electricity wastage, to inform energy efficiency actions.
- Load profiling of critical loads (KW). These data can be provided from hourly interval metering.
- Longitudinal benchmarking to provide visibility into long-term building's energy trends, including tracking whole building energy savings after implementation of energy efficiency projects
- Tracking fuel costs that help with reconciliation of electricity billing costs and identify variances in cost versus actual consumption.

A simple level of analysis can usually be carried out in-house. Notifications, such as basic alerts and stock recommendations, are provided to the facility manager, and standard monthly or quarterly reports are provided to executives. Actions such as scheduling and controls for main sub-metered end-uses and equipment can be conducted in-house. As needed, infrequent vendor support would be required for recalibration of meters, software upgrades, and the like.

EIS Package Tier 1 (Entry)

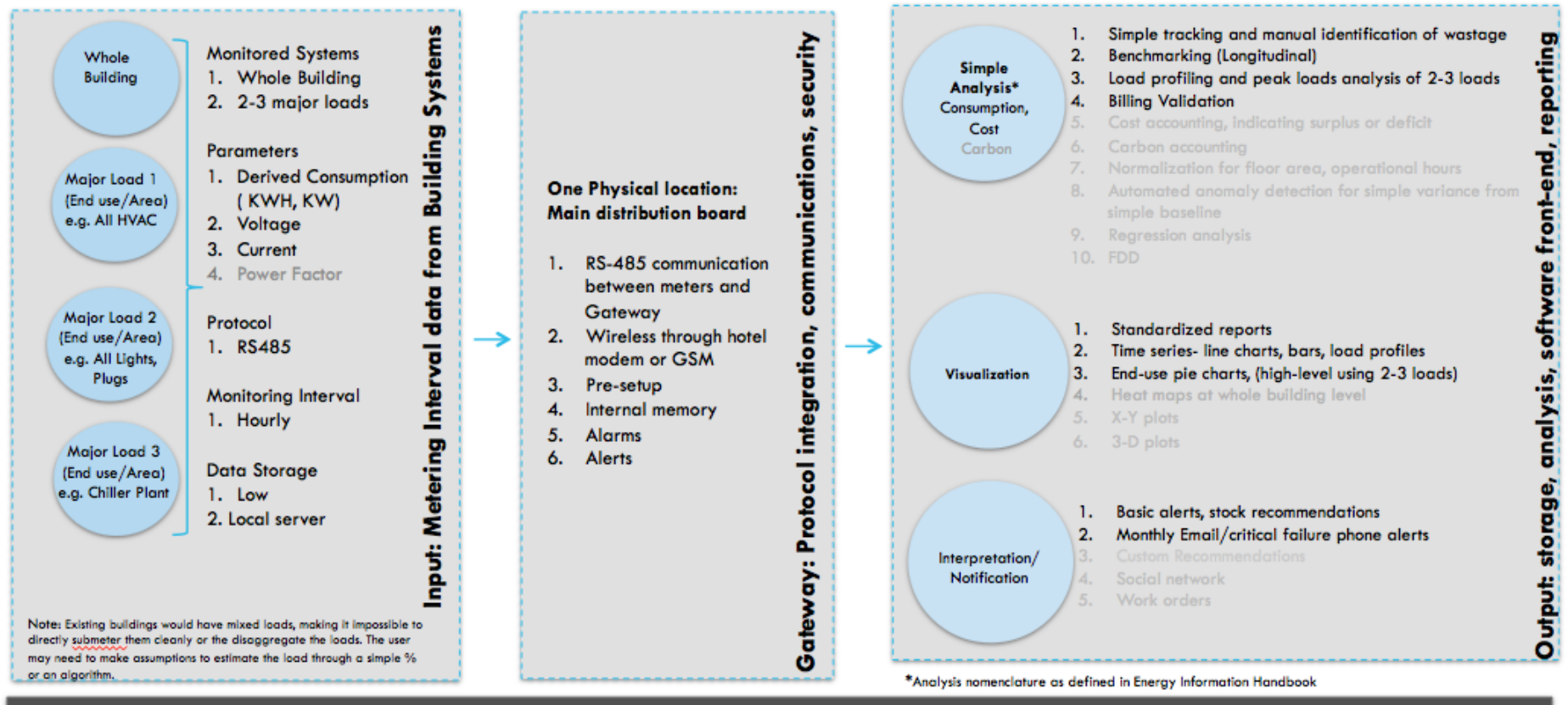


Figure 11: High-level overview of a Hotel EIS Tier 1 Entry Package. The left column recommends a picklist of loads, the middle column defines a simple communication gateway, and the right column provides the analysis and visualization functionality.

6.2 EIS Tier 2, Advanced package

Audience: The Tier 2, Advanced-Level package (Figure 12) is targeted toward hotel owners and managers who have a relatively higher awareness and interest in energy efficiency and carbon accounting to spend investment dollars and staff resources commensurately.

This is a more complex, granular package than the Entry level. The Advanced Package consolidates data from a few more interval meters and provides deeper visibility and analytics in terms of when, how much, and where electricity is being consumed and wasted. All other things being equal, this enables higher energy savings than the Entry-level package—up to ~10 percent—since it takes advantage of more extensive sub-metering to better pinpoint the reasons for use and waste. The cost is correspondingly higher because of additional metering requirement, higher functionality software, and/or ongoing vendor software services (analysis, data storage) cost, although the increase in number of points can potentially bring down the cost of the per point metering cost.

Data requirements: Interval data is recommended from up to 7 to 8 end-uses or major areas. Their selection is based on the picklist of loads. Beyond Entry level metering, other points such as outdoor lighting, air handlers, elevators, and major areas such as common areas and guest room blocks are recommended. Additional user-supplied information is also suggested to be configured into the EIS, such as operating schedules, building/ zone square feet areas, and designed occupancy (number of rooms, event schedule) that allow for normalization to provide superior analytics.

Visualization and Functionality: Six to ten advanced visualization screens are recommended. In addition to the charts provided in the EIS Basic package, there are charts depicting cost accounting, carbon accounting, and heat maps, as well as end-use pies, as illustrated in figures 13 and 14. In addition to the functionality and benefits of the Tier 1 Entry package, the Tier 2 Advanced package provides the following:

- Higher granularity and visibility into energy consumption (KWh) and load profiling (kW) of up to 7 to 8 major loads. Integration with additional user-provided data makes the Advanced EIS package a powerful tool to provide simple baselines that can be normalized (e.g., for floor area and operational hours) and can identify when and where the energy saving opportunities are with respect to time and the load category (i.e., scheduling, changes in load profile).
- Cross-sectional benchmarking with respect to a peer group such as a portfolio or other similar hotels. Benchmarking provides comparative information that reveals the need for improvement in energy performance; helps set energy targets; prioritizes energy efficiency projects, and tracks progress toward those targets.
- Cost accounting in terms of reporting electricity costs against the budget, indicating a surplus or deficit.
- Sustainability/GHG tracking, by providing carbon accounting analysis and reports.

Built-in notifications, such as email/phone alerts and some custom recommendations are also recommended in the Advanced package. The usability could be a conveniently handled through a hybrid model of in-house analysis and specific vendor-provided services. Vendor services can include support, training, and upgrade; as well as cost of any data storage in the cloud based or vendor-site local server.

EIS Package Tier 2 (Advanced)

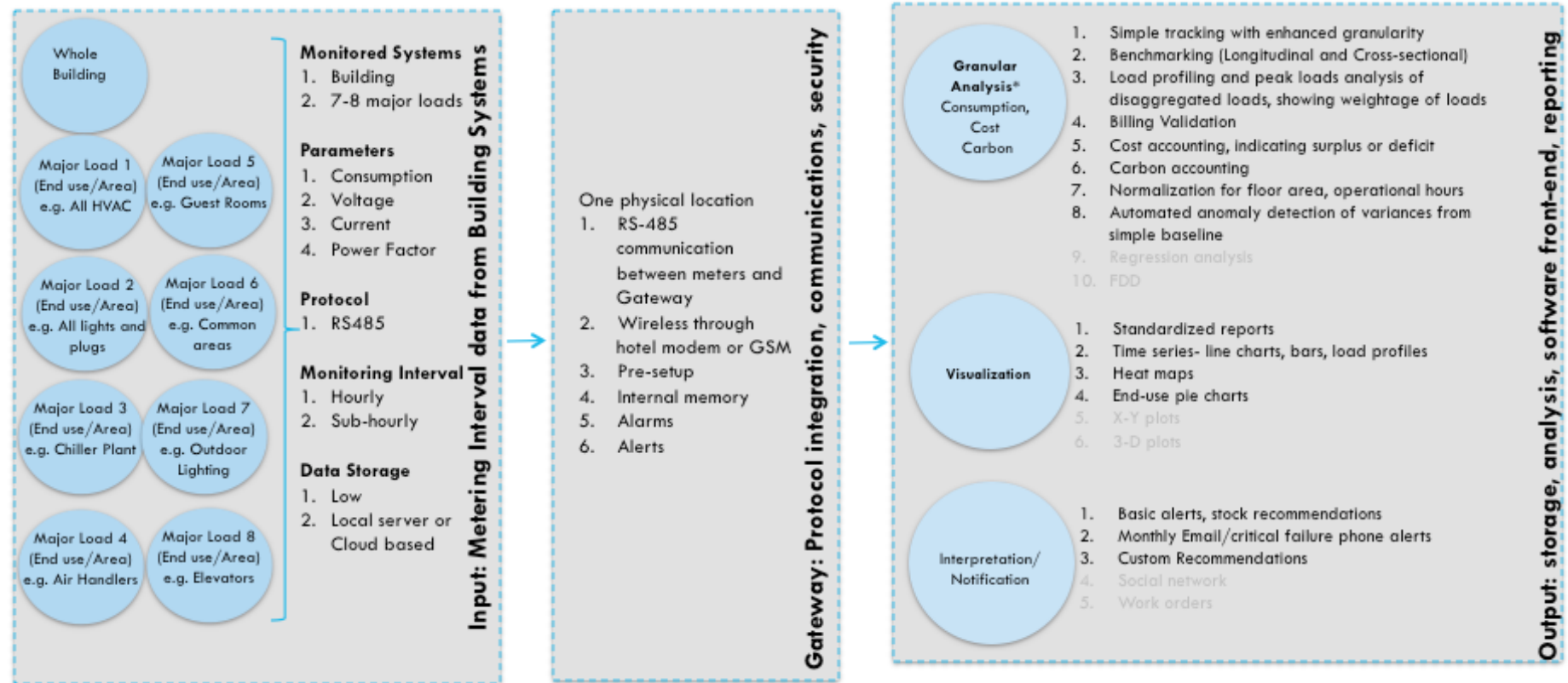


Figure 12: High-level overview of a Hotel EIS Tier 2 Advanced Package. The left column recommends a picklist of loads, the middle column defines a simple communication gateway, and the right column provides the analysis and visualization functionality.

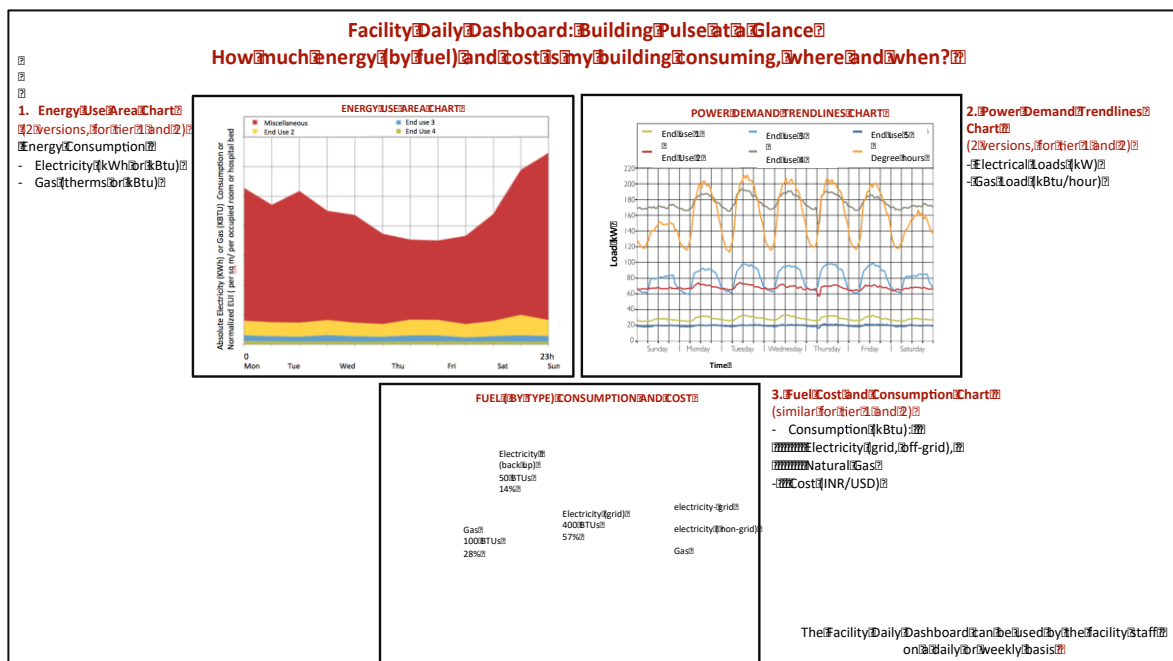


Figure 13: High-level overview of daily/weekly dashboard, “Building Pulse at a Glance” for facilities staff

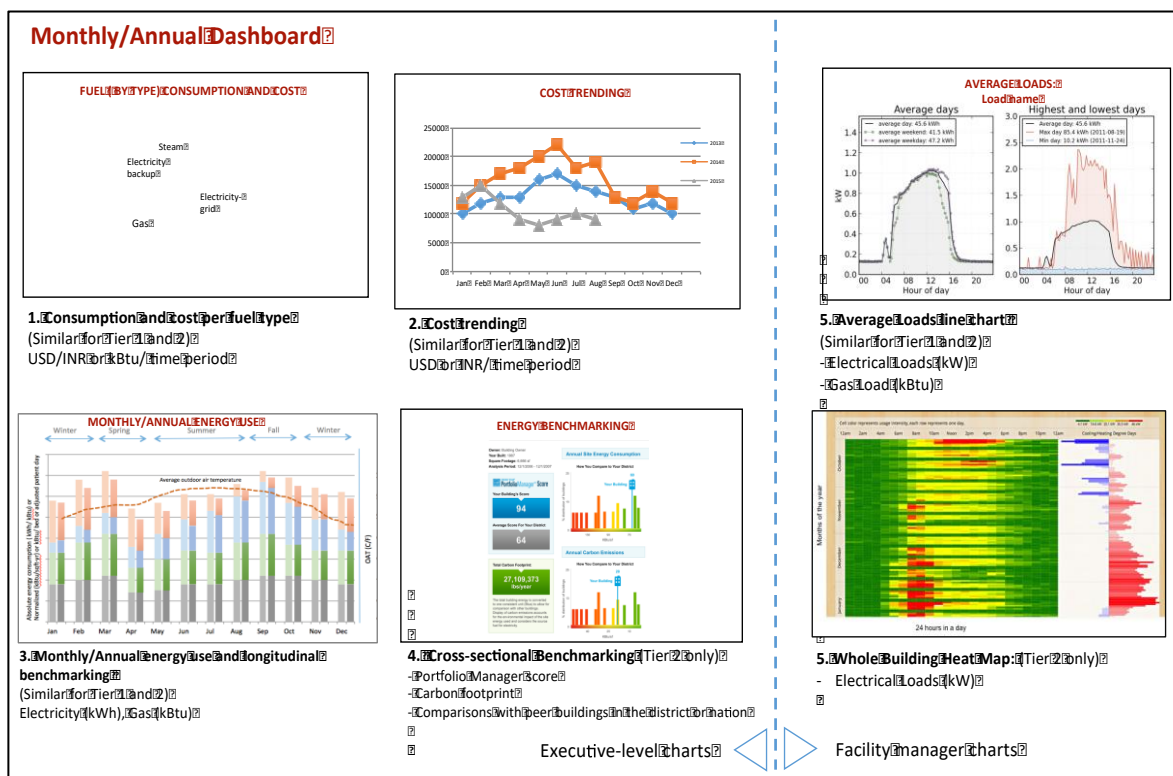


Figure 14: High-level overview of Monthly/ Annual dashboard for both executives and facilities staff

6.3 Trade-offs between functionality and greater adoption

There is a trade-off between functionality and broader adoption: Figure 15 shows the range of organizational and technical factors addressed in the two tiers of EIS packages. The tiered EIS-in-a-box packages do not provide the accuracy, granularity, and customized features of expensive custom energy information and management systems (EMIS) solutions used by some large buildings and portfolios. However, their value is in the 80-20 solution, i.e., they provide adequate functionality for energy savings in small- and medium-sized facilities, simplicity of use, actionable dashboards, and cost-effectiveness through first and operational cost reduction. We believe that the Tier 1 Entry EIS is a minimum requirement for all hotel facilities, and the Tier 2 Advanced EIS is best practice within a limited cost.



Figure 15: The two tiers of EIS-in-a-box provide trade-off between functionality and broader adoptability

7. Results

In a context where \$60 billion is spent annually on wasted energy in U.S. buildings (Forbes, 2015), it is critical to manage building energy use to achieve up to 20 percent savings enabled by EIS (Granderson et al., 2016). A key technical response to help overcome the scale and adoption challenge for EIS is the development of packaged solutions that address the current pain points around specifying, procuring, installing, and using the EIS. Our packaged EIS-in-a-box solution provides the following benefits:

- **Cost-effectiveness**, by helping to reduce first costs (i.e., transactional costs, hardware costs) as well as ongoing operational costs. This can help ease the sales cycle for vendors (especially for new or tough-to-penetrate markets) and make procurement and operations cost-effective for users. To test our assumptions around reduction of transaction costs, we conducted vendor

interviews and found that packaged EIS can help bring down cost by up to 30 percent of custom solutions (Singh et al., 2017).

- **Scalability**, by development of an optimum EIS that takes into account the specificity of the hotel sector rather than being too generic (all types of buildings) or too specific (customized on a per-building basis); also by engineering EIS packages to accommodate heterogeneity across hotel buildings. Its replicability across the organization's building portfolio and scalability establishes a broader market applicability of the systems. These EIS hotels packages are intended to be relevant across a range of facilities by recognizing and accommodating the heterogeneity within this target segment.
- **Simplicity**, through ease of use, procurement, and installation. EIS packages are engineered for the uncomplicated integration of the three main components (i.e., meters, gateways, and software) into preset configurations as two recommended tiered EIS-in-a-box packages. The uncluttered hardware and streamlined data architecture and cadence helps curtail unnecessary data management. Technical simplification of products and their usability is a real need and a path toward scalable deployment.
- **Actionability in the organization**, through alignment with business drivers and metrics that are relevant across various levels of organizational decision making. For the facilities staff in the basement checking the near-time, daily, or weekly pulse, it enables data-derived actions in the facilities operations. For executives, valuable energy information presented with the relevant energy and sustainability metrics at quarterly/annual time frames can help inform facility and energy investments within their larger decision-making framework.

The EIS package technical requirements that were developed in this work provide details for the three components—metering, gateways, and software/user interface—that are integrated into two tiers. The recommendations are specific, but allow flexibility in the prioritized selection of points for energy monitoring, reporting, and granularity of data acquisition, analysis, and actionable information display.

Energy Information Systems packages significantly improve on the business-as-usual practice for energy measurement of post-facto utility bill information, or for single-point-in-time information such as spot measurements, site gauges, manufacturer specifications, and assumptions. The core EIS information is intended to provide knowledge that is actionable across various types of decision makers in an organization, as well as across organizations in the offices sector. The packages rely on buy-in across various stakeholders, including upper management commitment for investment in these packaged solutions, as well as early involvement of IT staff to help overcome security, data maintenance, big data management, and installation hurdles. The packages rely upon training of in-house staff while leveraging the technical skills of systems integrators as necessary in the process. In this way, the design of the packages considers the EIS vendor, integrator, and client organization's facilities and IT staff as crucial partners in the successful installation of an EIS and persistent savings through its use. Finally, advantages such as standardization of packages also may help with interoperability and security standards in the near future.

While these EIS packages do not provide all the features available through more complex, custom-built EIS solutions, they represent a cost-effective option for stakeholders interested in increasing their property's energy efficiency who might otherwise be unable to practically leverage the technology. While sacrificing some of the granularity or accuracy, the value of the EIS-in-a-box is derived through their ease of use and adoption. The sector-specific packages can be a factor in market transformation that could allow building owners and managers to easily procure, install, and operate a system to monitor their energy usage, identify areas for improvement and cost savings, and encourage market adoption of the technology.

While this paper focuses on the hotel sector, our methodology spans three building sectors (hospitals, hotels, and offices) across two countries that are among the world's top energy consumers, namely the United States and India. Vendor interviews have been conducted to study typical transaction costs and project cost savings from EIS-in-a box packages across these building typologies. These packages are being field-demonstrated in six buildings in India and a key next step is to assess the efficacy of the packages and the effectiveness of data-driven decision making through the packages.

In conclusion, the research question we investigated was: How can we effectively design building energy information system packages that can overcome technological and market barriers to attain significant energy savings for high-consuming building sectors? Through streamlining and integrating the metering, communication, and visualization components of an EIS, it is possible to significantly reduce transaction costs, provide a relevant amount data for energy-based decision making, and drive significant energy savings through these adaptable and adoptable packaged EIS-in-a box solutions.

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